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Ministry of Works and Housing
Ghana Revenue Authority

GHANA RENTAL MARKET TAXATION SYSTEM (GRMTS)

FUNCTIONAL SPECIFICATION DOCUMENT

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1. Executive Summary

1.1 Purpose of This Document

This Functional Specification Document (FSD) provides a comprehensive and detailed description of the Ghana Rental Market Taxation System (GRMTS). It serves as the primary reference for all stakeholders involved in the design, development, testing, deployment, and operation of the system. This document captures the complete functional requirements, business rules, user interface specifications, and system behaviors necessary to build a system that meets the needs of the Ghana Revenue Authority, Ministry of Works and Housing, landlords, tenants, and other stakeholders.

The intended audience for this document includes:

- Government stakeholders and policy makers who need to understand system capabilities
- Project managers and business analysts responsible for requirements management
- Software architects and developers who will implement the system
- Quality assurance teams who will validate the system against these specifications
- Training teams who will develop user documentation and training materials
- Operations teams who will support the system post-deployment

1.2 Problem Statement

Ghana's rental housing market faces several critical challenges that negatively impact both government revenue collection and tenant welfare:

1.2.1 Tax Evasion in the Rental Sector

The rental income tax compliance rate in Ghana is estimated at less than 15%. With approximately 90% of Ghanaians living in rental accommodations and an estimated 2.5 million rental contracts nationwide, the tax leakage represents a significant loss of government revenue. Conservative estimates suggest annual tax losses exceeding GHS 800 million from the residential rental sector alone.

Contributing factors to this tax evasion include:

- Lack of mandatory contract registration requirements
- Cash-based transactions that leave no audit trail
- Limited enforcement capacity at the district level
- No systematic method to identify landlords and their rental income
- Complex tax filing requirements that discourage voluntary compliance

1.2.2 Excessive Advance Rent Demands

Landlords in Ghana commonly require tenants to pay rent advances of 1-3 years upfront before occupancy. This practice creates severe financial hardship for tenants who must accumulate

substantial sums before securing housing. A typical 2-bedroom apartment in Accra renting at GHS 2,000 per month with a 2-year advance requirement demands GHS 48,000 upfront - an amount that exceeds the annual income of many Ghanaian workers.

The consequences of excessive advance demands include:

- Reduced housing accessibility for low and middle-income families
- Increased informal lending at predatory interest rates
- Tenants trapped in substandard housing due to inability to save for moves
- Economic inefficiency as large sums remain locked in non-productive use
- Social inequality as housing access becomes wealth-dependent

1.2.3 Lack of Market Transparency

The absence of reliable rental market data creates information asymmetry that disadvantages tenants. Without access to comparable rental rates, tenants cannot effectively negotiate fair prices or identify overpriced properties. This opacity also hampers urban planning, policy formulation, and economic analysis of the housing sector.

1.2.4 Limited Tenant Protections

Current mechanisms for resolving landlord-tenant disputes are slow, expensive, and inaccessible to most tenants. The formal court system is overburdened, and many tenants are unaware of their legal rights. This power imbalance enables exploitation through arbitrary evictions, illegal fee collection, and poor property maintenance.

1.3 Proposed Solution

The Ghana Rental Market Taxation System (GRMTS) is a comprehensive digital platform designed to address these challenges through technology-enabled regulation, automated tax collection, and market transparency. The system will fundamentally transform how rental transactions are conducted in Ghana.

1.3.1 Core Solution Components

Mandatory Digital Contract Registration: All rental agreements must be registered through GRMTS to be legally valid and enforceable. Unregistered contracts will not be recognized by courts, police, or government agencies. This creates a powerful incentive for compliance while building a comprehensive database of rental activity.

Automatic Tax Withholding: When tenants make rent payments through the platform, the system automatically calculates and withholds the applicable withholding tax (8% for registered landlords, 15% for unregistered). Tax is remitted directly to GRA, eliminating the need for landlords to file separately and ensuring complete collection.

Advance Payment Limits: The system enforces maximum advance payment periods based on property type - 6 months for residential properties and 12-24 months for commercial properties. Contracts exceeding these limits cannot be registered, making them unenforceable.

Market Rent Database: Aggregated and anonymized rental data provides public access to fair market rent information by location and property type. This empowers tenants to negotiate effectively and helps landlords price competitively.

Digital Dispute Resolution: Built-in mechanisms for filing and resolving disputes provide accessible and affordable alternatives to court proceedings. Cases are assigned to qualified rent officers who can mediate and issue binding decisions.

Risk-Based Enforcement: Automated algorithms analyze data from multiple sources to identify potential violations and prioritize enforcement actions. Inspectors receive case assignments with risk scores and supporting evidence.

1.3.2 Key System Features

The GRMTS platform includes the following major features:

- Multi-channel access: Web portal, mobile application, and USSD for feature phones
- Identity verification through integration with National Identification Authority
- Property registration with ownership verification through Lands Commission
- Secure payment processing via Mobile Money (MTN, Vodafone, AirtelTigo) and bank transfer
- Automated tax certificate generation with QR code verification
- Real-time compliance monitoring dashboards for GRA officers
- Mobile-enabled inspector workflow with GPS tracking and evidence capture
- Anonymous tip reporting for suspected violations
- Comprehensive audit trail for all transactions and actions

1.4 Expected Benefits

1.4.1 Benefits for Government

Benefit Category	Description
Increased Tax Revenue	Projected 400% increase in rental income tax collection within 3 years through automatic withholding and expanded compliance base
Reduced Enforcement Costs	Risk-based targeting reduces wasted inspection resources by focusing on high-probability violations
Economic Data	Comprehensive rental market data supports urban planning, housing policy, and economic analysis
Formalization	Brings informal rental sector into regulated economy, enabling additional policy interventions
Reduced Corruption	Digital processes with audit trails reduce opportunities for corrupt practices in tax collection and enforcement

1.4.2 Benefits for Landlords

Benefit Category	Description
Simplified Compliance	Tax automatically withheld and remitted - no separate filing required for rental income
Tax Certificates	Automatic generation of withholding certificates for annual tax returns
Legal Protection	Registered contracts provide stronger legal standing in disputes and eviction proceedings
Tenant Verification	Access to tenant rental history and verification of identity
Payment Security	Digital payments reduce risks associated with cash handling

1.4.3 Benefits for Tenants

Benefit Category	Description
Reduced Advance Burden	Legal limits on advance payments reduce financial barriers to housing access
Price Transparency	Market rent data helps identify fair prices and strengthens negotiating position
Dispute Resolution	Accessible platform for filing complaints and resolving disputes without expensive litigation
Payment Records	Digital payment history serves as rental reference for future housing applications
Legal Protections	Registered contracts ensure tenant rights are documented and enforceable

1.5 Project Scope Summary

This functional specification covers the complete GRMTS platform including:

- User registration and identity verification for all user types
- Property registration and ownership verification
- Contract creation, tenant confirmation, and lifecycle management
- Payment processing with automatic tax withholding
- Tax certificate generation and verification
- Compliance monitoring and risk-based enforcement
- Inspector case management and field operations
- Dispute filing and resolution workflows
- Market rent data collection and public access
- USSD interface for feature phone users
- Administrative functions and system configuration
- Reporting and analytics for all stakeholder types

1.6 Success Metrics

The GRMTS project will be evaluated against the following key performance indicators:

Metric	Year 1 Target	Year 3 Target	Year 5 Target
Registered Landlords	50,000	250,000	500,000
Registered Properties	100,000	750,000	2,000,000
Active Contracts	150,000	1,000,000	2,500,000
Tax Revenue (GHS millions)	200	800	1,500
Contract Registration Rate	30%	70%	90%
Digital Payment Rate	80%	95%	99%
Dispute Resolution Time (days)	45	30	21

2. Introduction

2.1 Background and Problem Statement

In Ghana, 90% of the population lives in rental premises. The rental market faces critical challenges that have persisted for decades:

- Landlords commonly require tenants to prepay rent for up to two years in advance, creating severe financial stress
- The rental market remains largely unregulated, leading to widespread tax evasion
- Unfair practices go unchecked due to lack of formal contract registration
- Government loses significant tax revenue from untracked rental income
- Tenants have limited legal recourse without formal contracts
- No centralized data exists for market analysis and policy decisions

Previous attempts to improve rental tax compliance through voluntary registration have yielded limited results. The GRMTS represents a paradigm shift from voluntary compliance to mandatory digital registration with automatic tax collection.

2.2 Solution Overview

The Ghana Rental Market Taxation System (GRMTS) is a centralized digital system for registering all rental contracts with the following key capabilities:

- Mandatory digital contract registration (unregistered contracts are legally invalid)
- Automatic 8% withholding tax on all rent payments
- Real-time monitoring and enforcement by GRA officers
- Market transparency through aggregated rent data
- Mobile app for landlords and tenants with Ghana Card verification
- Web portal for GRA officers, inspectors, and administrators
- Digital signature capture for contract authentication
- Mobile money payment integration (MTN, Vodafone, AirtelTigo)
- Ghana Card OCR scanning for identity verification
- Push notification support for real-time alerts

2.3 System Architecture Overview

GRMTS implements a strict platform separation architecture:

2.3.1 Mobile Application (React Native/Expo)

The mobile app is exclusively for Landlords and Tenants. It provides:

- User registration with Ghana Card verification and OTP
- Property registration and management (Landlords)

- Digital contract creation with ID scanning using OCR
- Digital signature capture for contracts
- Contract confirmation and signing (Tenants)
- Mobile money payment processing
- Payment history and receipt generation
- Tax certificate download (Landlords)
- Push notifications support

2.3.2 Web Portal (React)

The web portal is exclusively for GRA Officers, Inspectors, and Administrators. It provides:

- Role-based dashboard with analytics
- Contract monitoring and search
- Payment tracking and tax reports
- Compliance monitoring and scoring
- Inspector case management
- Penalty issuance (GRA Supervisor)
- Audit management
- Report generation and export
- User administration

2.3.3 Backend Services (Node.js/Express)

The backend provides RESTful APIs for both platforms:

- JWT-based authentication with role-based access control
- PostgreSQL database with Sequelize ORM
- Document scanning and OCR integration
- Payment gateway integration for mobile money
- Tax calculation engine (8% withholding)
- Push notification service via Expo

2.4 Legal and Regulatory Framework

The GRMTS operates within the following legal framework:

2.4.1 Primary Legislation

Rent Act, 1963 (Act 220): Governs landlord-tenant relationships, establishes tenant protections and landlord obligations.

Income Tax Act, 2015 (Act 896): Defines rental income as taxable and establishes the 8% withholding tax requirement on rent payments.

Revenue Administration Act, 2016 (Act 915): Provides GRA's administrative powers for tax collection and enforcement.

Data Protection Act, 2012 (Act 843): Governs personal data handling and privacy requirements.

Electronic Transactions Act, 2008 (Act 772): Provides legal validity for digital contracts and electronic signatures.

2.4.2 Tax Regulations

All rental payments processed through GRMTS are subject to 8% withholding tax at source:

- Tax is automatically calculated and deducted from every payment
- Tax is remitted directly to GRA
- Landlords receive tax certificates for their records
- Tax certificates can be downloaded from the mobile app

2.5 Stakeholder Analysis

The following stakeholders have significant interest in the GRMTS:

Stakeholder	Interest	Platform	Impact
Landlords	Simplified tax compliance, legal protection	Mobile App	High
Tenants	Fair rental terms, payment tracking	Mobile App	High
Ghana Revenue Authority	Tax collection, compliance monitoring	Web Portal	High
Ministry of Finance	Revenue increase, policy data	Reports	Medium
Rent Control Dept	Dispute resolution, monitoring	Web Portal	Medium
Mobile Money Providers	Transaction volume	Integration	Medium
NIA	Ghana Card verification services	Integration	Low

2.6 Document Conventions

This document uses the following conventions:

- **SHALL:** Indicates a mandatory requirement that must be implemented
- **SHOULD:** Indicates a recommended feature that is strongly encouraged
- **MAY:** Indicates an optional feature that can be implemented
- Requirement IDs follow the format: [MODULE]-[TYPE]-[NUMBER] (e.g., USR-FR-001)
- **[Mobile]:** Feature available on mobile app only (Landlord/Tenant)

- [Web]: Feature available on web portal only (GRA/Inspector/Admin)
- [Both]: Feature available on both platforms
- GHS refers to Ghanaian Cedis throughout this document

2.7 Glossary of Terms

Term	Definition
GRMTS	Ghana Rental Market Taxation System - the complete solution
GRA	Ghana Revenue Authority - primary tax collection agency
Ghana Card	National identification card issued by NIA, required for registration
TIN	Taxpayer Identification Number - unique tax identifier
Digital Address	Ghana Post GPS address format (XX-XXX-XXXX)
Withholding Tax	8% tax deducted at source from all rental payments
OCR	Optical Character Recognition - extracts text from Ghana Card scans
OTP	One-Time Password - verification code sent via SMS
Mobile Money	Electronic payment via MTN MoMo, Vodafone Cash, or AirtelTigo Money
JWT	JSON Web Token - secure authentication mechanism
Expo	React Native development platform for mobile app

3. Scope and Objectives

3.1 Project Scope

3.1.1 In-Scope Features - Mobile Application

The following features are implemented in the React Native/Expo mobile application:

- User registration with Ghana Card verification and OTP
- Property registration and management (Landlords only)
- Digital contract creation with tenant ID scanning
- Ghana Card OCR data extraction with confidence scoring
- Manual correction support for OCR data
- Digital signature capture (full-screen signature pad)
- Contract confirmation workflow (Tenants)
- Mobile money payment integration (MTN, Vodafone, AirtelTigo)
- Payment history and receipt generation
- Tax certificate download (Landlords)
- Push notifications support
- Profile management

3.1.2 In-Scope Features - Web Portal

The following features are implemented in the React web portal:

- Role-restricted access (GRA/Inspector/Admin only)
- Dashboard with analytics and KPIs
- Property management views and search
- Contract monitoring and filtering
- Payment tracking and reporting
- Compliance reports and scoring
- Inspector case management
- Mobile app redirect notice for landlords/tenants attempting web login
- Data export capabilities (CSV, PDF)
- User administration (Admin only)

3.1.3 In-Scope Features - Backend Services

The following backend capabilities are implemented:

- RESTful API with JWT authentication
- PostgreSQL database with Sequelize ORM
- User management with role-based access control

- Property registration and management
- Contract creation and management
- Payment processing with automatic 8% tax calculation
- Document scanning and OCR integration
- Simulated external service integrations (NIA, GRA, Mobile Money)

3.1.4 Out-of-Scope Features

The following features are explicitly excluded from the current implementation:

- Direct bank account integration (mobile money only)
- International payment processing
- Property valuation services
- Legal case management for courts
- Integration with mortgage providers
- Multi-currency support (GHS only)
- Offline mode for mobile app
- USSD interface for feature phones
- Property listing marketplace
- Tenant credit scoring

3.2 Platform Access Matrix

GRMTS enforces strict platform separation to ensure appropriate access:

Role	Web Portal	Mobile App
LANDLORD	NOT ALLOWED	FULL ACCESS
TENANT	NOT ALLOWED	FULL ACCESS
GRA_OFFICER	FULL ACCESS	NOT AVAILABLE
GRA_SUPERVISOR	FULL ACCESS	NOT AVAILABLE
INSPECTOR	FULL ACCESS	NOT AVAILABLE
ADMIN	FULL ACCESS	NOT AVAILABLE
SYSTEM_ADMIN	FULL ACCESS	NOT AVAILABLE

3.3 Business Objectives

The GRMTS project aims to achieve the following measurable objectives:

3.3.1 Revenue Objectives

- Increase rental income tax collection through automatic 8% withholding
- Reduce tax leakage by making unregistered contracts legally invalid

- Provide transparent tax certificates for landlord compliance

3.3.2 Compliance Objectives

- Register all rental contracts digitally with both party signatures
- Process rent payments exclusively through the platform
- Achieve high tenant confirmation rate through mobile verification
- Enforce advance rent limits through system validation

3.3.3 Operational Objectives

- Maintain high system availability for business operations
- Process mobile money payments within 30 seconds
- Generate tax certificates automatically upon payment completion
- Enable GRA officers to monitor compliance in real-time

4. User Roles and Access Control

4.1 Role Overview

The GRMTS supports seven distinct user roles, strictly separated by platform:

Role Name	Platform	Primary Responsibilities
LANDLORD	Mobile	Register properties, create contracts, scan tenant IDs, sign contracts, view payments, download tax certificates
TENANT	Mobile	Confirm contracts, scan own ID, sign contracts, make mobile money payments, view payment history
GRA_OFFICER	Web	View all contracts and payments, view tax reports, monitor compliance, export reports, search users
GRA_SUPERVISOR	Web	All GRA_OFFICER permissions plus: issue penalties, manage audits, manage officers
INSPECTOR	Web	View assigned cases, update case status, upload evidence, submit inspection reports
ADMIN	Web	Full access to all system features, user management, system configuration
SYSTEM_ADMIN	Web	Full access including technical configuration and database management

4.2 Mobile App Roles

4.2.1 LANDLORD

Property owners who use the mobile app to manage their rental properties and contracts.

Registration Requirements:

- Valid Ghana Card number (verified through NIA integration)
- Active mobile phone number (for OTP verification)
- Email address
- Personal information (name, date of birth, gender)
- Digital address for correspondence

Permissions:

Permission	Description
register_property	Add new rental properties with details, photos, and digital address
create_contract	Create new rental contracts for registered properties
scan_tenant_id	Scan tenant's Ghana Card using camera with OCR extraction
sign_contract	Provide digital signature using signature pad
view_own_contracts	View all contracts for owned properties
view_own_payments	View payment history for own contracts

download_tax_certificate	Download 8% withholding tax certificates for tax filing
update_profile	Update personal profile and notification preferences

Landlord User Journey:

- 1. Download mobile app and register with Ghana Card
- 2. Complete OTP verification (demo: use code 123456)
- 3. Register rental property with digital address
- 4. Create contract by scanning tenant's Ghana Card
- 5. Review OCR-extracted data and add contract terms
- 6. Provide digital signature
- 7. Share confirmation code with tenant
- 8. Monitor payments and download tax certificates

4.2.2 TENANT

Individuals who rent properties and use the mobile app to confirm contracts and make payments.

Registration Requirements:

- Valid Ghana Card number (verified through NIA integration)
- Active mobile phone number (for OTP verification)
- Email address
- Personal information (name, date of birth, gender)

Permissions:

Permission	Description
confirm_contract	Review and confirm pending contracts using confirmation code
scan_own_id	Scan own Ghana Card for verification during contract confirmation
sign_contract	Provide digital signature to complete contract activation
make_payment	Process rent payments via MTN, Vodafone, or AirtelTigo mobile money
view_own_contracts	View contracts where user is the tenant
view_own_payments	View personal payment history and download receipts
update_profile	Update personal profile and notification preferences

Tenant User Journey:

- 1. Download mobile app and register with Ghana Card
- 2. Complete OTP verification (demo: use code 123456)

- 3. Receive confirmation code from landlord
- 4. Enter code to view pending contract details
- 5. Review contract terms and scan own Ghana Card
- 6. Provide digital signature to confirm contract
- 7. Make rent payments via mobile money
- 8. View payment history and download receipts

4.3 Web Portal Roles

4.3.1 GRA_OFFICER

Ghana Revenue Authority staff members who monitor rental contracts and tax compliance. This is a read-only role focused on monitoring and reporting.

Permissions:

Permission	Description
view_all_contracts	Search and view all registered contracts in the system
view_all_payments	View all payment transactions and history
view_tax_reports	Access tax collection reports and summaries
view_compliance_data	Monitor compliance metrics and statistics
export_reports	Export data to CSV/PDF formats for analysis
search_landlords	Search and view landlord profiles and properties
search_tenants	Search and view tenant profiles and contracts

4.3.2 GRA_SUPERVISOR

Senior GRA staff with elevated permissions for enforcement actions and audit management.

Additional Permissions (beyond GRA_OFFICER):

Permission	Description
issue_penalty	Assess and issue penalties for non-compliance violations
manage_audits	Create, assign, and manage audit cases for investigation
manage_officers	Manage GRA officer accounts and work assignments

4.3.3 INSPECTOR

Field officers who investigate compliance cases and conduct property inspections. Access is limited to assigned cases only.

Permissions:

Permission	Description
view_assigned_cases	View only inspection cases assigned to this inspector

update_case_status	Update status of assigned investigation cases
upload_evidence	Upload photos and documents as case evidence
submit_inspection_report	Submit formal inspection findings report
view_property_details	View property information for assigned cases only
view_contract_details	View contract details for assigned cases only

4.3.4 ADMIN / SYSTEM_ADMIN

Administrators with full system access. ADMIN handles user and operational management, while SYSTEM_ADMIN handles technical configuration.

Permissions (Wildcard *):

- All GRA_OFFICER permissions
- All GRA_SUPERVISOR permissions
- All INSPECTOR permissions
- User account management (create, modify, deactivate)
- System configuration and settings
- Audit log access and review
- Database management operations
- Integration configuration
- Reference data management

4.4 Role-Permission Matrix

Complete permission matrix showing all permissions by role:

Permission	Landlord	Tenant	GRA Off	GRA Sup	Inspector	Admin
Register Property	Yes	-	-	-	-	Yes
Create Contract	Yes	-	-	-	-	Yes
Confirm Contract	-	Yes	-	-	-	Yes
Scan Documents	Yes	Yes	-	-	-	-
Sign Contract	Yes	Yes	-	-	-	-
Make Payment	-	Yes	-	-	-	-
View All Contracts	-	-	Yes	Yes	-	Yes
View All Payments	-	-	Yes	Yes	-	Yes
Issue Penalties	-	-	-	Yes	-	Yes
Manage Cases	-	-	-	-	Yes	Yes
Export Reports	-	-	Yes	Yes	-	Yes
Tax Certificate	Yes	-	-	-	-	Yes

System Config	-	-	-	-	-	Yes
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5. Functional Requirements

This section provides detailed functional requirements for each module of the GRMTS. Requirements are organized by functional area and include unique identifiers for traceability.

Requirement priorities are defined as:

- Must Have (M): Essential requirements that must be implemented for system launch
- Should Have (S): Important requirements that should be implemented if possible
- Could Have (C): Desirable requirements for future consideration
- Won't Have (W): Requirements explicitly excluded from current scope

5.1 User Management Module

The User Management Module handles all aspects of user registration, authentication, profile management, and access control. It integrates with external identity verification services to ensure user authenticity.

5.1.1 User Registration

The system shall support self-registration for external users (landlords and tenants) with identity verification through the National Identification Authority.

Functional Requirements - Registration:

Req ID	Requirement Description	Priority	Status
USR-FR-001	System shall provide a registration form capturing: email, phone number, password, and role selection (Landlord/Tenant)	Must Have	Approved
USR-FR-002	System shall validate email format and check for existing accounts with same email	Must Have	Approved
USR-FR-003	System shall validate Ghana phone number format (10 digits starting with 0)	Must Have	Approved
USR-FR-004	System shall enforce password complexity: minimum 8 characters, at least one uppercase, one lowercase, one number	Must Have	Approved
USR-FR-005	System shall send OTP to provided phone number within 30 seconds of registration submission	Must Have	Approved
USR-FR-006	System shall allow 3 OTP retry attempts before requiring new OTP generation	Must Have	Approved
USR-FR-007	OTP shall expire after 10 minutes from generation	Must Have	Approved
USR-FR-008	System shall collect Ghana Card number after phone verification	Must Have	Approved
USR-FR-009	System shall verify Ghana Card with NIA and retrieve: full name, date of birth, gender, photo, address	Must Have	Approved
USR-FR-010	System shall display NIA-retrieved information for user confirmation before account creation	Must Have	Approved
USR-FR-011	System shall reject registration if Ghana Card is already associated with another account	Must Have	Approved
USR-FR-012	For landlord registration, system shall prompt for TIN or offer to generate new TIN through GRA integration	Must Have	Approved

USR-FR-013	System shall collect and verify TIN with GRA database	Should Have	Approved
USR-FR-014	System shall collect bank account or mobile money number for landlord settlement	Must Have	Approved
USR-FR-015	System shall send welcome SMS and email upon successful registration	Should Have	Approved
USR-FR-016	Corporate registration shall require additional fields: company name, registration number, authorized representative details	Must Have	Approved
USR-FR-017	Corporate registration shall require upload of: certificate of incorporation, board resolution, representative ID	Must Have	Approved
USR-FR-018	Corporate accounts shall require manual verification before activation (within 48 hours)	Must Have	Approved

Registration Workflow:

Step 1: User accesses registration page and enters email, phone, and password

Step 2: System validates input and sends OTP to phone number

Step 3: User enters OTP received via SMS

Step 4: System verifies OTP and prompts for role selection

Step 5: User selects Landlord or Tenant role

Step 6: User enters Ghana Card number

Step 7: System verifies Ghana Card with NIA and displays retrieved information

Step 8: User confirms information is correct

Step 9: For Landlords: User enters TIN (or requests new TIN) and settlement account

Step 10: System creates account and sends confirmation

5.1.2 User Authentication

The system shall provide secure authentication mechanisms supporting multiple factors.

Functional Requirements - Authentication:

Req ID	Requirement Description	Priority	Status
USR-FR-020	System shall authenticate users with email/phone and password	Must Have	Approved
USR-FR-021	System shall lock account after 5 consecutive failed login attempts for 30 minutes	Must Have	Approved
USR-FR-022	System shall support password reset via OTP sent to registered phone	Must Have	Approved
USR-FR-023	System shall require current password for password change when logged in	Must Have	Approved

USR-FR-024	System shall maintain session for 30 minutes of inactivity before automatic logout	Must Have	Approved
USR-FR-025	System shall support 'Remember Me' option extending session to 30 days	Should Have	Approved
USR-FR-026	System shall invalidate all sessions when password is changed	Must Have	Approved
USR-FR-027	System shall log all authentication attempts with timestamp, IP address, and result	Must Have	Approved
USR-FR-028	System shall support two-factor authentication via OTP for sensitive operations	Should Have	Approved
USR-FR-029	System shall prevent concurrent sessions from different devices (optional setting)	Could Have	Approved
USR-FR-030	System shall provide 'Forgot Password' flow with identity verification	Must Have	Approved

5.1.3 User Profile Management

Users shall be able to view and update their profile information subject to verification requirements.

Functional Requirements - Profile:

Req ID	Requirement Description	Priority	Status
USR-FR-040	System shall display user profile with all registered information	Must Have	Approved
USR-FR-041	System shall allow update of contact information (email, phone) with re-verification	Must Have	Approved
USR-FR-042	System shall allow update of settlement account details with verification	Must Have	Approved
USR-FR-043	System shall NOT allow modification of Ghana Card number (requires new registration)	Must Have	Approved
USR-FR-044	System shall allow users to upload profile photo	Should Have	Approved
USR-FR-045	System shall allow users to set notification preferences (SMS, email, push)	Must Have	Approved
USR-FR-046	System shall allow users to set preferred language (English, Akan, Ewe, etc.)	Should Have	Approved
USR-FR-047	System shall display compliance score for landlords	Must Have	Approved
USR-FR-048	System shall display verification status badges (identity verified, TIN verified, etc.)	Should Have	Approved
USR-FR-049	System shall allow landlords to add/remove settlement accounts	Must Have	Approved
USR-FR-050	System shall maintain profile change history with timestamps	Must Have	Approved

5.1.4 Corporate Sub-User Management

Corporate accounts shall be able to create and manage sub-user accounts for property managers and staff.

Functional Requirements - Sub-Users:

Req ID	Requirement Description	Priority	Status
USR-FR-060	Corporate primary user shall be able to create sub-user accounts	Must Have	Approved
USR-FR-061	Sub-user creation shall require: name, email, phone, Ghana Card (optional)	Must Have	Approved
USR-FR-062	System shall send invitation to sub-user with temporary password	Must Have	Approved
USR-FR-063	Primary user shall be able to assign specific properties to sub-users	Must Have	Approved
USR-FR-064	Primary user shall be able to set permission levels for sub-users	Must Have	Approved
USR-FR-065	Sub-user permissions shall not exceed primary user permissions	Must Have	Approved
USR-FR-066	Primary user shall be able to deactivate sub-user accounts	Must Have	Approved
USR-FR-067	System shall log all actions taken by sub-users with user identification	Must Have	Approved
USR-FR-068	Primary user shall be able to view activity log for all sub-users	Should Have	Approved
USR-FR-069	Corporate accounts shall support up to 50 sub-users	Should Have	Approved
USR-FR-070	Sub-users shall be able to reset their own passwords	Must Have	Approved

5.2 Property Management Module

The Property Management Module enables landlords to register their rental properties, upload documentation, and manage property availability. It integrates with external services for address validation and ownership verification.

5.2.1 Property Registration

Landlords shall register properties with comprehensive details to enable contract creation and compliance monitoring.

Functional Requirements - Property Registration:

Req ID	Requirement Description	Priority	Status
PRP-FR-001	System shall require Ghana Post digital address for all properties	Must Have	Approved
PRP-FR-002	System shall validate digital address format (XX-XXX-XXXX)	Must Have	Approved
PRP-FR-003	System shall verify digital address with Ghana Post GPS service	Should Have	Approved
PRP-FR-004	System shall require selection of property category (Residential/Commercial)	Must Have	Approved
PRP-FR-005	System shall require selection of property type based on category	Must Have	Approved
PRP-FR-006	For residential properties, system shall capture number of bedrooms (0-10+)	Must Have	Approved
PRP-FR-007	For residential properties, system shall capture number of bathrooms (0-10+)	Must Have	Approved
PRP-FR-008	System shall capture floor area in square meters (optional)	Should Have	Approved
PRP-FR-009	System shall capture year built (optional)	Should Have	Approved
PRP-FR-010	System shall capture furnishing status (Unfurnished/Semi-Furnished/Fully Furnished)	Must Have	Approved

Req ID	Requirement Description	Priority	Status
PRP-FR-011	System shall capture amenities as multi-select (parking, security, generator, water tank, AC, etc.)	Should Have	Approved
PRP-FR-012	System shall require at least one exterior photo of property	Must Have	Approved
PRP-FR-013	System shall require at least one interior photo per bedroom	Should Have	Approved
PRP-FR-014	Photo uploads shall support JPEG, PNG formats up to 10MB each	Must Have	Approved
PRP-FR-015	System shall extract GPS coordinates from photo EXIF data when available	Should Have	Approved
PRP-FR-016	System shall allow manual GPS coordinate entry if EXIF not available	Should Have	Approved
PRP-FR-017	System shall require selection of ownership type (Freehold/Leasehold/Family/Estate/Corporate)	Must Have	Approved
PRP-FR-018	System shall require upload of ownership documentation	Must Have	Approved

PRP-FR-019	Ownership documents shall support PDF, JPEG, PNG formats up to 20MB	Must Have	Approved
PRP-FR-020	System shall generate unique property code upon registration	Must Have	Approved

Req ID	Requirement Description	Priority	Status
PRP-FR-021	Property code format shall be: [REGION]-[DISTRICT]-[DIGITAL_ADDRESS]-[SEQUENCE]	Must Have	Approved
PRP-FR-022	System shall set initial property status to 'Pending Verification'	Must Have	Approved
PRP-FR-023	System shall attempt ownership verification with Lands Commission integration	Should Have	Approved
PRP-FR-024	System shall update status to 'Verified' upon successful ownership verification	Must Have	Approved
PRP-FR-025	If automated verification fails, system shall queue for manual review	Must Have	Approved
PRP-FR-026	Individual landlords shall be limited to 10 properties (enforced at registration)	Must Have	Approved
PRP-FR-027	System shall notify landlord of property status changes via SMS and email	Should Have	Approved
PRP-FR-028	System shall capture region and district from digital address lookup	Must Have	Approved
PRP-FR-029	System shall allow optional neighborhood/area name entry	Should Have	Approved
PRP-FR-030	System shall store full street address if provided	Should Have	Approved

Property Types - Residential:

Code	Type Name	Description
R-SR	Single Room	Single room in shared house, shared facilities
R-SC	Self-Contained	Single room with private bathroom and/or kitchen
R-CH	Chamber and Hall	Bedroom plus separate living room, may share facilities
R-1B	1-Bedroom Apartment	Self-contained unit with one bedroom
R-2B	2-Bedroom Apartment	Self-contained unit with two bedrooms
R-3B	3-Bedroom Apartment	Self-contained unit with three bedrooms
R-4B	4+ Bedroom Apartment	Self-contained unit with four or more bedrooms
R-HS	Standalone House	Detached house with private compound
R-VL	Villa/Mansion	Luxury detached property with extensive grounds
R-TH	Townhouse	Multi-story unit sharing walls with neighbors
R-CMP	Compound House	Multiple units within a shared compound

Property Types - Commercial:

Code	Type Name	Description
C-SHP	Shop/Retail	Retail space in commercial area or shopping center
C-OFF	Office Space	Office accommodation in commercial building
C-WHS	Warehouse	Storage and distribution facility
C-IND	Industrial	Manufacturing or industrial facility
C-RST	Restaurant/Food Service	Food service establishment premises
C-HTL	Hotel/Hospitality	Accommodation for hospitality business
C-MXD	Mixed Use	Combined residential and commercial use
C-LND	Land Only	Vacant land for commercial use

5.2.2 Property Modification

Landlords shall be able to update property information subject to restrictions.

Functional Requirements - Property Updates:

Req ID	Requirement Description	Priority	Status
PRP-FR-040	System shall allow update of property amenities at any time	Must Have	Approved
PRP-FR-041	System shall allow update of furnishing status at any time	Must Have	Approved
PRP-FR-042	System shall allow addition of new photos at any time	Must Have	Approved
PRP-FR-043	System shall allow removal of photos if minimum requirements still met	Should Have	Approved
PRP-FR-044	System shall NOT allow change of digital address (requires new registration)	Must Have	Approved
PRP-FR-045	System shall NOT allow change of property type if active contracts exist	Must Have	Approved
PRP-FR-046	Change of ownership documentation shall trigger re-verification	Must Have	Approved
PRP-FR-047	System shall maintain history of all property changes with timestamps	Must Have	Approved
PRP-FR-048	System shall allow landlord to mark property as unavailable	Must Have	Approved
PRP-FR-049	Marking property unavailable shall not affect active contracts	Must Have	Approved
PRP-FR-050	System shall allow reactivation of unavailable properties	Must Have	Approved

5.2.3 Property Verification

The system shall verify property ownership and validity through automated and manual processes.

Functional Requirements - Verification:

Req ID	Requirement Description	Priority	Status
PRP-FR-060	System shall attempt automated verification through Lands Commission API	Should Have	Approved

PRP-FR-061	Automated verification shall check: property exists, owner name matches	Should Have	Approved
PRP-FR-062	If automated verification succeeds, status shall change to Verified within 24 hours	Should Have	Approved
PRP-FR-063	If automated verification fails, case shall be queued for manual review	Must Have	Approved
PRP-FR-064	Manual verification queue shall be visible to District Officers	Must Have	Approved
PRP-FR-065	District Officers shall be able to approve or reject pending properties	Must Have	Approved
PRP-FR-066	Rejection shall require reason selection from predefined list	Must Have	Approved
PRP-FR-067	Landlord shall be notified of verification outcome via SMS and email	Must Have	Approved
PRP-FR-068	Rejected properties shall be able to be resubmitted with corrections	Must Have	Approved
PRP-FR-069	Properties pending verification for >30 days shall trigger escalation	Should Have	Approved
PRP-FR-070	Verified properties shall display verification badge in search results	Should Have	Approved

5.2.4 Multi-Unit Property Support

The system shall support registration of properties with multiple rentable units.

Functional Requirements - Multi-Unit:

Req ID	Requirement Description	Priority	Status
PRP-FR-080	System shall allow marking a property as multi-unit during registration	Must Have	Approved
PRP-FR-081	Multi-unit properties shall specify total number of units	Must Have	Approved
PRP-FR-082	Each unit shall have a unique unit identifier (e.g., Apt 1, Unit A)	Must Have	Approved
PRP-FR-083	Units may have different property types (e.g., mix of 1-bed and 2-bed)	Should Have	Approved
PRP-FR-084	Contracts shall be created at unit level, not property level	Must Have	Approved
PRP-FR-085	System shall track occupancy status per unit	Must Have	Approved
PRP-FR-086	Property dashboard shall show unit-level occupancy summary	Should Have	Approved
PRP-FR-087	Ownership verification shall apply to entire property, not individual units	Must Have	Approved
PRP-FR-088	Each unit shall be able to have its own photos in addition to common areas	Should Have	Approved
PRP-FR-089	Market rent data shall be calculated per unit type within multi-unit properties	Should Have	Approved

5.3 Contract Management Module

The Contract Management Module handles the complete lifecycle of rental contracts from creation through termination. It enforces business rules around advance payments and ensures tenant confirmation for contract validity.

5.3.1 Contract Creation

Landlords shall create contracts for their registered properties with specified terms and conditions.

Functional Requirements - Contract Creation:

Req ID	Requirement Description	Priority	Status
CTR-FR-001	System shall allow landlords to create contracts for verified properties only	Must Have	Approved
CTR-FR-002	System shall prevent contract creation for properties with active contracts (unless multi-unit)	Must Have	Approved
CTR-FR-003	System shall require tenant identification: existing user OR phone number + Ghana Card	Must Have	Approved
CTR-FR-004	If tenant not registered, system shall create pending tenant record	Must Have	Approved
CTR-FR-005	System shall capture contract start date (cannot be more than 90 days in past)	Must Have	Approved
CTR-FR-006	System shall capture contract end date (minimum 1 month, maximum 60 months from start)	Must Have	Approved
CTR-FR-007	System shall calculate contract duration and display in months	Must Have	Approved
CTR-FR-008	System shall capture monthly rent amount in GHS	Must Have	Approved
CTR-FR-009	System shall enforce minimum rent of GHS 50 per month	Should Have	Approved
CTR-FR-010	System shall capture number of advance months required	Must Have	Approved

Req ID	Requirement Description	Priority	Status
CTR-FR-011	System shall enforce maximum advance months based on property type	Must Have	Approved
CTR-FR-012	For residential properties: maximum 6 months advance	Must Have	Approved
CTR-FR-013	For commercial shop/office/mixed: maximum 12 months advance	Must Have	Approved
CTR-FR-014	For commercial warehouse/industrial: maximum 24 months advance	Must Have	Approved
CTR-FR-015	System shall capture security deposit amount (optional)	Must Have	Approved
CTR-FR-016	System shall enforce maximum security deposit of 3 months rent	Must Have	Approved
CTR-FR-017	System shall capture service charge amount if applicable (optional)	Should Have	Approved
CTR-FR-018	System shall allow selection of payment frequency (Monthly/Quarterly/Annually)	Should Have	Approved
CTR-FR-019	System shall require upload of signed contract document (PDF or image)	Must Have	Approved
CTR-FR-020	Contract document upload shall support up to 20MB file size	Must Have	Approved

Req ID	Requirement Description	Priority	Status
CTR-FR-021	System shall display calculated total initial payment (advance + deposit + service)	Must Have	Approved
CTR-FR-022	System shall display applicable tax rate and estimated tax amount	Must Have	Approved
CTR-FR-023	System shall generate unique contract number upon creation	Must Have	Approved
CTR-FR-024	Contract number format: CTR-[YEAR]-[SEQUENCE]	Must Have	Approved
CTR-FR-025	System shall generate 6-digit OTP confirmation code for tenant	Must Have	Approved
CTR-FR-026	System shall set contract status to PENDING_CONFIRMATION upon creation	Must Have	Approved
CTR-FR-027	System shall send SMS to tenant with contract details and confirmation code	Must Have	Approved
CTR-FR-028	System shall send email to tenant if email address available	Should Have	Approved
CTR-FR-029	Confirmation code shall expire after 7 days	Must Have	Approved
CTR-FR-030	System shall allow landlord to resend confirmation code (max 3 times)	Must Have	Approved

5.3.2 Tenant Confirmation

Tenants must confirm contracts for them to become active and legally valid.

Functional Requirements - Confirmation:

Req ID	Requirement Description	Priority	Status
CTR-FR-040	System shall display pending contracts to tenant upon login	Must Have	Approved
CTR-FR-041	System shall show complete contract details: property, landlord, terms, rent, deposit	Must Have	Approved
CTR-FR-042	System shall display market rent comparison for property type and location	Should Have	Approved
CTR-FR-043	System shall display warning if rent exceeds 90th percentile for area	Should Have	Approved
CTR-FR-044	System shall allow tenant to view uploaded contract document	Must Have	Approved
CTR-FR-045	Tenant shall enter confirmation code to confirm contract	Must Have	Approved
CTR-FR-046	System shall verify confirmation code matches and is not expired	Must Have	Approved
CTR-FR-047	Upon valid confirmation, contract status shall change to ACTIVE	Must Have	Approved
CTR-FR-048	System shall record confirmation timestamp and tenant IP address	Must Have	Approved
CTR-FR-049	System shall notify landlord of contract confirmation via SMS and email	Must Have	Approved
CTR-FR-050	Tenant shall be able to object to contract instead of confirming	Must Have	Approved

Req ID	Requirement Description	Priority	Status
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CTR-FR-051	Objection shall require selection of reason from predefined list	Must Have	Approved
CTR-FR-052	Objection reasons: Incorrect rent, Incorrect terms, Not my contract, Excess advance, Other	Must Have	Approved
CTR-FR-053	Objection shall require comments describing the issue	Must Have	Approved
CTR-FR-054	Upon objection, contract status shall change to DISPUTED	Must Have	Approved
CTR-FR-055	System shall create dispute case automatically upon objection	Must Have	Approved
CTR-FR-056	System shall notify landlord of objection with reason	Must Have	Approved
CTR-FR-057	Contracts not confirmed within 7 days shall expire automatically	Must Have	Approved
CTR-FR-058	Expired contracts shall change status to EXPIRED_UNCONFIRMED	Must Have	Approved
CTR-FR-059	Landlord shall be notified when confirmation period expires	Must Have	Approved
CTR-FR-060	Landlord may create new contract after expiry	Must Have	Approved

5.3.3 Contract Lifecycle Management

The system shall manage contracts through their complete lifecycle including renewal and termination.

Functional Requirements - Lifecycle:

Req ID	Requirement Description	Priority	Status
CTR-FR-070	System shall send reminder 60 days before contract expiry	Should Have	Approved
CTR-FR-071	System shall send reminder 30 days before contract expiry	Must Have	Approved
CTR-FR-072	Landlord shall be able to initiate contract renewal	Must Have	Approved
CTR-FR-073	Renewal shall create new contract linked to original	Must Have	Approved
CTR-FR-074	Renewal shall pre-populate property and tenant from original contract	Should Have	Approved
CTR-FR-075	Renewed contracts shall require tenant confirmation	Must Have	Approved
CTR-FR-076	Either party shall be able to initiate termination with notice	Must Have	Approved
CTR-FR-077	Termination shall require selection of reason from predefined list	Must Have	Approved
CTR-FR-078	Termination shall require specification of termination date	Must Have	Approved
CTR-FR-079	Minimum notice period for termination: 30 days	Must Have	Approved
CTR-FR-080	Other party shall be notified of termination request	Must Have	Approved

Req ID	Requirement Description	Priority	Status
CTR-FR-081	Contracts shall automatically change to EXPIRED status on end date	Must Have	Approved
CTR-FR-082	System shall maintain complete contract history including all status changes	Must Have	Approved
CTR-FR-083	System shall generate contract summary document with QR code	Should Have	Approved

CTR-FR-084	QR code shall link to verification page showing contract status	Should Have	Approved
CTR-FR-085	Third parties shall be able to verify contract status via QR code without login	Should Have	Approved
CTR-FR-086	Contract modification after confirmation shall require agreement from both parties	Should Have	Approved
CTR-FR-087	System shall track all contract modifications with before/after values	Must Have	Approved
CTR-FR-088	Rent increases shall be limited to once per 12-month period	Should Have	Approved
CTR-FR-089	Rent increases exceeding 20% shall trigger review flag	Should Have	Approved
CTR-FR-090	System shall prevent deletion of contracts (soft delete only)	Must Have	Approved

Contract Status Flow:

DRAFT -> PENDING_CONFIRMATION -> ACTIVE -> EXPIRED

DRAFT -> PENDING_CONFIRMATION -> DISPUTED -> ACTIVE (if resolved) or CANCELLED

DRAFT -> PENDING_CONFIRMATION -> EXPIRED_UNCONFIRMED

ACTIVE -> TERMINATED (early termination by either party)

ACTIVE -> RENEWED (new contract created, original marked RENEWED)

5.4 Payment Processing Module

The Payment Processing Module enables tenants to make rent payments through the platform with automatic tax withholding and settlement to landlords. It integrates with multiple payment providers for broad accessibility.

5.4.1 Payment Initiation

Tenants shall be able to initiate payments for their active contracts through various channels.

Functional Requirements - Payment Initiation:

Req ID	Requirement Description	Priority	Status
PAY-FR-001	System shall display active contracts with outstanding balances to tenant	Must Have	Approved
PAY-FR-002	System shall show payment due date and days overdue if applicable	Must Have	Approved
PAY-FR-003	System shall allow selection of contract for payment	Must Have	Approved
PAY-FR-004	System shall pre-populate amount with monthly rent or outstanding balance	Should Have	Approved
PAY-FR-005	System shall allow tenant to modify payment amount (minimum: 1 month rent)	Must Have	Approved
PAY-FR-006	System shall allow payment of multiple months in single transaction	Should Have	Approved
PAY-FR-007	System shall calculate payment breakdown: gross, tax (8%), net to landlord	Must Have	Approved
PAY-FR-008	Tax calculation: flat 8% withholding tax on all rent payments	Must Have	Approved
PAY-FR-009	Tax formula: $\text{tax_amount} = \text{gross_amount} * 0.08$	Must Have	Approved
PAY-FR-010	Net to landlord formula: $\text{net_amount} = \text{gross_amount} - \text{tax_amount}$	Must Have	Approved

Req ID	Requirement Description	Priority	Status
PAY-FR-011	Platform fee: not applicable in current implementation	Should Have	Approved
PAY-FR-012	System shall display clear breakdown before payment confirmation	Must Have	Approved
PAY-FR-013	System shall show available payment methods to tenant	Must Have	Approved
PAY-FR-014	Available methods: MTN Mobile Money, Vodafone Cash, AirtelTigo Money	Must Have	Approved
PAY-FR-015	System shall remember tenant's preferred payment method	Should Have	Approved
PAY-FR-016	System shall pre-populate phone number from user profile for mobile money	Should Have	Approved
PAY-FR-017	Tenant shall confirm phone number before mobile money payment	Must Have	Approved
PAY-FR-018	System shall generate unique payment reference for each transaction	Must Have	Approved

PAY-FR-019	Payment reference format: PAY-[YEAR]-[SEQUENCE]	Must Have	Approved
PAY-FR-020	System shall set initial payment status to PENDING	Must Have	Approved

5.4.2 Payment Processing

The system shall process payments through integrated payment providers with appropriate error handling.

Functional Requirements - Processing:

Req ID	Requirement Description	Priority	Status
PAY-FR-030	System shall initiate payment request to selected payment provider	Must Have	Approved
PAY-FR-031	For mobile money: provider shall prompt user on their phone to authorize	Must Have	Approved
PAY-FR-032	System shall poll for payment status every 5 seconds	Should Have	Approved
PAY-FR-033	System shall timeout payment after 5 minutes if not completed	Must Have	Approved
PAY-FR-034	System shall display real-time status updates to user during processing	Should Have	Approved
PAY-FR-035	Upon success, system shall update payment status to COMPLETED	Must Have	Approved
PAY-FR-036	Upon failure, system shall update payment status to FAILED with reason	Must Have	Approved
PAY-FR-037	System shall record provider reference number for reconciliation	Must Have	Approved
PAY-FR-038	System shall record processing timestamps: initiated, completed/failed	Must Have	Approved
PAY-FR-039	System shall handle partial payments (if supported by provider)	Should Have	Approved
PAY-FR-040	System shall retry failed payments up to 3 times before final failure	Should Have	Approved

5.4.3 Payment Settlement

Successfully collected payments shall be settled to landlords after tax remittance.

Functional Requirements - Settlement:

Req ID	Requirement Description	Priority	Status
PAY-FR-050	System shall calculate net settlement amount: gross - tax - platform fee	Must Have	Approved
PAY-FR-051	System shall queue successful payments for settlement batch	Must Have	Approved
PAY-FR-052	Settlement batches shall run twice daily (6:00 AM and 6:00 PM)	Should Have	Approved
PAY-FR-053	System shall initiate settlement transfer to landlord's registered account	Must Have	Approved

PAY-FR-054	Settlement shall be via mobile money or bank transfer based on landlord preference	Must Have	Approved
PAY-FR-055	System shall record settlement status: PENDING, COMPLETED, FAILED	Must Have	Approved
PAY-FR-056	System shall notify landlord via SMS when settlement is initiated	Must Have	Approved
PAY-FR-057	System shall notify landlord via SMS when settlement is completed	Must Have	Approved
PAY-FR-058	Failed settlements shall be retried in next batch	Must Have	Approved
PAY-FR-059	Settlements failed 3 times shall be flagged for manual review	Must Have	Approved
PAY-FR-060	System shall maintain complete audit trail of all settlements	Must Have	Approved

5.4.4 Payment Receipts

The system shall generate receipts for all completed payments.

Functional Requirements - Receipts:

Req ID	Requirement Description	Priority	Status
PAY-FR-070	System shall generate digital receipt immediately upon successful payment	Must Have	Approved
PAY-FR-071	Receipt shall include: date, amount breakdown, reference, property, parties	Must Have	Approved
PAY-FR-072	Receipt shall include QR code for verification	Should Have	Approved
PAY-FR-073	System shall send receipt via SMS (shortened URL)	Must Have	Approved
PAY-FR-074	System shall send receipt via email if address available	Should Have	Approved
PAY-FR-075	Tenants shall be able to download receipt as PDF from dashboard	Must Have	Approved
PAY-FR-076	Landlords shall be able to view all receipts for their properties	Must Have	Approved
PAY-FR-077	Third parties shall be able to verify receipt via QR code	Should Have	Approved
PAY-FR-078	System shall maintain receipt history indefinitely	Must Have	Approved
PAY-FR-079	Receipts shall be searchable by date, contract, amount	Should Have	Approved

5.4.5 Payment Error Handling

The system shall handle payment errors gracefully with appropriate user feedback.

Functional Requirements - Errors:

Req ID	Requirement Description	Priority	Status
PAY-FR-080	System shall display user-friendly error messages for all failure scenarios	Must Have	Approved
PAY-FR-081	Insufficient funds: 'Your account balance is insufficient. Please top up and try again.'	Must Have	Approved

PAY-FR-082	Wrong PIN: 'Incorrect PIN entered. Please try again.'	Must Have	Approved
PAY-FR-083	Timeout: 'Payment timed out. Please check your account and try again.'	Must Have	Approved
PAY-FR-084	Provider unavailable: 'Payment service temporarily unavailable. Please try later.'	Must Have	Approved
PAY-FR-085	System shall log all errors with full technical details for debugging	Must Have	Approved
PAY-FR-086	System shall provide option to retry failed payments	Must Have	Approved
PAY-FR-087	System shall provide option to try different payment method	Should Have	Approved
PAY-FR-088	Failed payments shall not affect contract standing	Must Have	Approved
PAY-FR-089	System shall prevent duplicate payments within 5 minutes for same contract	Must Have	Approved

5.5 Tax and Certificates Module

The Tax and Certificates Module manages automatic tax withholding, remittance to GRA, and generation of withholding certificates for landlords.

5.5.1 Tax Withholding

The system shall automatically calculate and withhold applicable taxes on all rent payments.

Functional Requirements - Withholding:

Req ID	Requirement Description	Priority	Status
TAX-FR-001	System shall apply flat 8% withholding tax on all rent payments	Must Have	Approved
TAX-FR-002	Tax is automatically calculated: $\text{tax_amount} = \text{gross_amount} * 0.08$	Must Have	Approved
TAX-FR-003	Tax is deducted at source before landlord receives funds	Must Have	Approved
TAX-FR-004	Example: GHS 1,000 rent = GHS 80 tax, GHS 920 to landlord	Must Have	Approved
TAX-FR-005	Tax rate is consistent regardless of landlord type or TIN status	Must Have	Approved
TAX-FR-006	System shall display tax breakdown during payment flow	Must Have	Approved
TAX-FR-007	Payment screen shall show: Gross Amount, Tax (8%), Net to Landlord	Must Have	Approved
TAX-FR-008	System shall record tax amount for each payment transaction	Must Have	Approved
TAX-FR-009	System shall maintain running total of tax withheld per landlord	Must Have	Approved
TAX-FR-010	System shall maintain running total of tax withheld per contract	Must Have	Approved

5.5.2 Tax Certificate Generation

The system shall generate tax withholding certificates for landlords to use for tax filing.

Functional Requirements - Certificates:

Req ID	Requirement Description	Priority	Status
TAX-FR-020	System shall generate monthly certificates by 5th business day of following month	Must Have	Approved
TAX-FR-021	Monthly certificate shall show all payments and tax withheld during month	Must Have	Approved
TAX-FR-022	System shall generate annual summary certificate by January 31st	Must Have	Approved
TAX-FR-023	Annual certificate shall show totals by property for the tax year	Must Have	Approved
TAX-FR-024	Certificate shall include unique certificate number	Must Have	Approved
TAX-FR-025	Certificate number format: CERT-[YEAR]-[MONTH]-[SEQUENCE]	Must Have	Approved
TAX-FR-026	Certificate shall include QR code linking to verification page	Must Have	Approved
TAX-FR-027	Certificate shall include landlord TIN and name	Must Have	Approved
TAX-FR-028	Certificate shall be downloadable as PDF	Must Have	Approved
TAX-FR-029	System shall notify landlord when new certificate is available	Should Have	Approved

TAX-FR-030	Landlord shall be able to request custom period certificate	Should Have	Approved
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5.5.3 Certificate Verification

Third parties shall be able to verify certificate authenticity through a public portal.

Functional Requirements - Verification:

Req ID	Requirement Description	Priority	Status
TAX-FR-040	System shall provide public verification portal (no login required)	Must Have	Approved
TAX-FR-041	Verification by certificate number and landlord TIN	Must Have	Approved
TAX-FR-042	Verification by QR code scan	Should Have	Approved
TAX-FR-043	Verification result shall show: certificate validity, period, total tax withheld	Must Have	Approved
TAX-FR-044	Verification shall not display detailed transaction data	Must Have	Approved
TAX-FR-045	System shall log all verification attempts for audit	Should Have	Approved
TAX-FR-046	Invalid certificate queries shall be flagged for review	Should Have	Approved
TAX-FR-047	Verification portal shall be available 24/7	Must Have	Approved

5.5.4 Tax Reporting to GRA

The system shall provide comprehensive tax reports to Ghana Revenue Authority.

Functional Requirements - GRA Reporting:

Req ID	Requirement Description	Priority	Status
TAX-FR-050	System shall generate daily tax collection summary for GRA	Must Have	Approved
TAX-FR-051	Daily report shall include: total collections, landlord count, transaction count	Must Have	Approved
TAX-FR-052	System shall generate monthly detailed tax report by region and district	Must Have	Approved
TAX-FR-053	System shall provide GRA dashboard showing real-time collection statistics	Must Have	Approved
TAX-FR-054	GRA officers shall be able to drill down from summary to individual transactions	Must Have	Approved
TAX-FR-055	System shall flag unusual patterns for GRA review	Should Have	Approved
TAX-FR-056	System shall export data in GRA-specified formats	Must Have	Approved
TAX-FR-057	System shall integrate with GRA tax administration system via API	Should Have	Approved

5.6 Compliance and Enforcement Module

The Compliance and Enforcement Module enables monitoring of rental market compliance, risk-based detection of violations, and management of inspection cases.

5.6.1 Compliance Scoring

The system shall calculate and maintain compliance scores for all landlords.

Functional Requirements - Compliance Score:

Req ID	Requirement Description	Priority	Status
ENF-FR-001	System shall calculate compliance score (0-100) for each landlord	Must Have	Approved
ENF-FR-002	Contract registration factor: 30% weight - % of properties with registered contracts	Must Have	Approved
ENF-FR-003	Payment timeliness factor: 25% weight - % of payments received on time	Must Have	Approved
ENF-FR-004	Tenant confirmation factor: 15% weight - % of contracts confirmed by tenants	Must Have	Approved
ENF-FR-005	Document completeness factor: 10% weight - ownership docs verified	Must Have	Approved
ENF-FR-006	Dispute history factor: 10% weight - deduction for lost disputes	Must Have	Approved
ENF-FR-007	Inspection results factor: 10% weight - deduction for violations	Must Have	Approved
ENF-FR-008	System shall recalculate score daily	Must Have	Approved
ENF-FR-009	Score changes >10 points shall trigger alert to landlord	Should Have	Approved
ENF-FR-010	Scores below 50 shall flag landlord as high risk	Must Have	Approved

5.6.2 Risk Detection

The system shall automatically identify potential violations through data analysis.

Functional Requirements - Risk Detection:

Req ID	Requirement Description	Priority	Status
ENF-FR-020	System shall calculate risk score for each registered property	Must Have	Approved
ENF-FR-021	Property registered but no contract >6 months: +25 risk points	Must Have	Approved
ENF-FR-022	GSM presence detected but no contract: +20 risk points	Should Have	Approved
ENF-FR-023	Utility consumption anomaly (high usage, no contract): +15 risk points	Should Have	Approved
ENF-FR-024	Anonymous tip received: +15 risk points per tip	Must Have	Approved
ENF-FR-025	Previous violations by landlord: +10 risk points	Must Have	Approved
ENF-FR-026	Contract irregularities detected: +10 risk points	Should Have	Approved
ENF-FR-027	Risk score >60 shall trigger automatic case creation	Must Have	Approved
ENF-FR-028	System shall prioritize cases by risk score	Must Have	Approved

ENF-FR-029	System shall exclude recently inspected properties (within 6 months)	Should Have	Approved
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5.6.3 Anonymous Reporting

The system shall enable anonymous reporting of suspected violations.

Functional Requirements - Anonymous Tips:

Req ID	Requirement Description	Priority	Status
ENF-FR-030	System shall provide anonymous tip reporting form (no login required)	Must Have	Approved
ENF-FR-031	Tip form shall capture: location, description, evidence (optional)	Must Have	Approved
ENF-FR-032	System shall attempt to match tip to registered property	Should Have	Approved
ENF-FR-033	Unmatched tips shall be queued for manual location identification	Must Have	Approved
ENF-FR-034	Tip submission shall return tracking number for status checks	Should Have	Approved
ENF-FR-035	Reporter may provide contact for follow-up (optional, anonymity preserved)	Should Have	Approved
ENF-FR-036	System shall prevent spam tips from same IP (max 5 per day)	Must Have	Approved
ENF-FR-037	Tips matching same property shall be consolidated	Should Have	Approved

5.6.4 Inspection Case Management

District Officers and Inspectors shall manage field inspection cases through the system.

Functional Requirements - Case Management:

Req ID	Requirement Description	Priority	Status
ENF-FR-040	System shall create cases automatically from risk detection	Must Have	Approved
ENF-FR-041	District Officers shall be able to create cases manually	Must Have	Approved
ENF-FR-042	Case shall include: property details, risk factors, evidence, history	Must Have	Approved
ENF-FR-043	District Officer shall assign cases to inspectors	Must Have	Approved
ENF-FR-044	Assignment shall consider inspector workload and location	Should Have	Approved
ENF-FR-045	Inspector shall receive notification of new assignment	Must Have	Approved
ENF-FR-046	Inspector shall schedule inspection date	Must Have	Approved
ENF-FR-047	System shall notify property owner of scheduled inspection (optional)	Should Have	Approved
ENF-FR-048	Inspector shall check in at location via GPS	Must Have	Approved
ENF-FR-049	GPS check-in shall verify inspector is within 100m of property	Should Have	Approved
ENF-FR-050	Inspector shall upload photos and evidence from mobile device	Must Have	Approved

Req ID	Requirement Description	Priority	Status
ENF-FR-051	Inspector shall complete inspection report form	Must Have	Approved
ENF-FR-052	Report form shall include structured findings and free-text notes	Must Have	Approved
ENF-FR-053	Report shall capture: occupancy status, tenant interview results, violations found	Must Have	Approved
ENF-FR-054	Inspector shall select outcome: Violation Confirmed, No Violation, Inconclusive	Must Have	Approved
ENF-FR-055	For violations, inspector shall specify violation type and severity	Must Have	Approved
ENF-FR-056	Completed report shall be submitted for supervisor review	Must Have	Approved
ENF-FR-057	District Officer shall review and approve/reject inspection reports	Must Have	Approved
ENF-FR-058	Approved violations shall trigger penalty assessment workflow	Must Have	Approved
ENF-FR-059	Case shall be closed with final outcome recorded	Must Have	Approved
ENF-FR-060	Closed cases shall update property and landlord risk profiles	Must Have	Approved

5.6.5 Penalty Assessment

The system shall support assessment and collection of penalties for violations.

Penalty Schedule:

Violation Type	First Offense	Second Offense	Third+ Offense
Unregistered Contract	100% annual rent	200% annual rent	300% + prosecution
Excess Advance	50% of excess	100% of excess	200% of excess
False Information	GHS 10,000	GHS 50,000	Prosecution
Cash Evasion	100% amount	200% amount	License revoke
Missing QR Display	GHS 5,000	GHS 20,000	Closure order

Req ID	Requirement Description	Priority	Status
ENF-FR-070	System shall calculate penalty amount based on violation type and history	Must Have	Approved
ENF-FR-071	GRA Officer shall approve penalty assessment before issuance	Must Have	Approved
ENF-FR-072	System shall generate penalty notice with payment instructions	Must Have	Approved
ENF-FR-073	System shall allow penalty payment through same payment channels	Must Have	Approved
ENF-FR-074	Unpaid penalties shall accrue interest at 2% per month	Should Have	Approved
ENF-FR-075	Penalties unpaid >90 days shall trigger escalation workflow	Must Have	Approved

5.7 Dispute Resolution Module

The Dispute Resolution Module provides mechanisms for landlords and tenants to file and resolve disputes through the platform.

5.7.1 Dispute Filing

Either party to a contract shall be able to file disputes for resolution.

Functional Requirements - Filing:

Req ID	Requirement Description	Priority	Status
DIS-FR-001	System shall allow landlords and tenants to file disputes for their contracts	Must Have	Approved
DIS-FR-002	Dispute filing shall require selection of dispute type	Must Have	Approved
DIS-FR-003	Types: Contract Terms, Payment, Habitability, Eviction, Deposit Return, Rent Increase, Other	Must Have	Approved
DIS-FR-004	Filing shall require detailed description of issue (minimum 50 characters)	Must Have	Approved
DIS-FR-005	Filing shall allow upload of supporting evidence (documents, photos)	Must Have	Approved
DIS-FR-006	Filing shall specify requested resolution	Should Have	Approved
DIS-FR-007	System shall generate unique dispute number	Must Have	Approved
DIS-FR-008	Dispute number format: DIS-[YEAR]-[SEQUENCE]	Must Have	Approved
DIS-FR-009	System shall set initial status to FILED	Must Have	Approved
DIS-FR-010	System shall notify other party of dispute filing	Must Have	Approved

5.7.2 Dispute Response

The responding party shall have opportunity to respond to disputes.

Functional Requirements - Response:

Req ID	Requirement Description	Priority	Status
DIS-FR-020	Respondent shall have 7 days to submit response	Must Have	Approved
DIS-FR-021	Response shall address each point raised in complaint	Should Have	Approved
DIS-FR-022	Response shall allow upload of counter-evidence	Must Have	Approved
DIS-FR-023	System shall change status to UNDER_REVIEW after response or after 7 days	Must Have	Approved
DIS-FR-024	Original filer may submit reply to response (within 3 days)	Should Have	Approved
DIS-FR-025	System shall limit rounds of response to 2 per party	Must Have	Approved

5.7.3 Dispute Resolution

Rent Officers shall review disputes and issue resolutions.

Functional Requirements - Resolution:

Req ID	Requirement Description	Priority	Status
DIS-FR-030	System shall assign disputes to Rent Officers based on district	Must Have	Approved
DIS-FR-031	Rent Officer shall review all submissions and evidence	Must Have	Approved
DIS-FR-032	Rent Officer may request additional information from either party	Should Have	Approved
DIS-FR-033	Rent Officer may schedule mediation session	Should Have	Approved
DIS-FR-034	Rent Officer shall issue written resolution with reasoning	Must Have	Approved
DIS-FR-035	Resolution may include: dismissed, awarded to complainant, awarded to respondent, compromise	Must Have	Approved
DIS-FR-036	Resolution may include monetary award up to GHS 100,000	Should Have	Approved
DIS-FR-037	System shall notify both parties of resolution	Must Have	Approved
DIS-FR-038	Either party may appeal resolution within 14 days	Must Have	Approved
DIS-FR-039	Appeals shall be referred to senior Rent Officer or Rent Control Board	Should Have	Approved
DIS-FR-040	System shall track dispute resolution time (target: 30 days)	Must Have	Approved

5.8 Market Rent Data Module

The Market Rent Data Module aggregates rental market information and provides public access to fair market rent data.

5.8.1 Data Aggregation

The system shall aggregate and analyze rental data from registered contracts.

Functional Requirements - Aggregation:

Req ID	Requirement Description	Priority	Status
MKT-FR-001	System shall aggregate rent data from active contracts monthly	Must Have	Approved
MKT-FR-002	Data shall be grouped by: region, district, neighborhood, property type, bedrooms	Must Have	Approved
MKT-FR-003	System shall calculate: average, median, min, max, 10th/90th percentiles	Must Have	Approved
MKT-FR-004	Minimum sample size of 5 contracts required for statistics display	Must Have	Approved
MKT-FR-005	Data shall exclude outliers (rents >3 standard deviations from mean)	Should Have	Approved
MKT-FR-006	System shall maintain 12 months of historical data for trends	Must Have	Approved
MKT-FR-007	Data shall be anonymized - no individual property/landlord/tenant identification	Must Have	Approved

5.8.2 Market Rent Checker

The public shall have access to a tool for checking typical rental prices.

Functional Requirements - Rent Checker:

Req ID	Requirement Description	Priority	Status
MKT-FR-020	System shall provide public Market Rent Checker (no login required)	Must Have	Approved
MKT-FR-021	User shall select: region (required), district (required)	Must Have	Approved
MKT-FR-022	User shall select: neighborhood (optional), property type (required)	Must Have	Approved
MKT-FR-023	For residential, user shall select number of bedrooms	Must Have	Approved
MKT-FR-024	System shall display: average rent, median rent, typical range	Must Have	Approved
MKT-FR-025	System shall display sample size and data recency	Must Have	Approved
MKT-FR-026	System shall display 12-month trend chart	Should Have	Approved
MKT-FR-027	User may enter specific rent amount for comparison	Should Have	Approved
MKT-FR-028	Comparison shall show percentile ranking of entered rent	Should Have	Approved
MKT-FR-029	If insufficient data, system shall show nearest comparable area	Should Have	Approved

5.9 Notifications Module

The Notifications Module manages all system communications to users across multiple channels.

5.9.1 Notification Types

The system shall support the following notification categories:

Event	Description	SMS	Email
Registration Complete	Welcome message after account creation	Yes	Yes
Contract Created	Landlord notified of contract creation	Yes	Yes
Contract Pending	Tenant notified to confirm contract	Yes	Yes
Contract Confirmed	Both parties notified of active contract	Yes	Yes
Payment Received	Landlord notified of payment receipt	Yes	Yes
Payment Completed	Tenant confirmation of successful payment	Yes	Yes
Payment Due Reminder	Reminder 7 days before payment due	Yes	No
Payment Overdue	Alert for overdue payments	Yes	Yes
Contract Expiring	Reminder 30/60 days before expiry	Yes	Yes
Tax Certificate Ready	Monthly/annual certificate available	Yes	Yes
Dispute Filed	Party notified of dispute against them	Yes	Yes
Dispute Resolved	Resolution notification	Yes	Yes
Inspection Scheduled	Property inspection notice	Yes	Yes
Penalty Issued	Penalty assessment notification	Yes	Yes

Functional Requirements - Notifications:

Req ID	Requirement Description	Priority	Status
NOT-FR-001	System shall send SMS notifications to registered phone number	Must Have	Approved
NOT-FR-002	System shall send email notifications to registered email address	Should Have	Approved
NOT-FR-003	System shall maintain in-app notification center	Must Have	Approved
NOT-FR-004	Users shall be able to configure notification preferences by type	Should Have	Approved
NOT-FR-005	SMS delivery shall be attempted within 60 seconds of event	Must Have	Approved
NOT-FR-006	Failed SMS shall be retried up to 3 times over 1 hour	Should Have	Approved
NOT-FR-007	System shall log all notification attempts and delivery status	Must Have	Approved
NOT-FR-008	Notifications shall support English and local languages (Akan, Ewe)	Should Have	Approved
NOT-FR-009	Critical notifications (payment, contract) shall not be opt-out	Must Have	Approved
NOT-FR-010	Users shall be able to mark in-app notifications as read	Must Have	Approved

6. User Interface Specifications

This section provides detailed specifications for each screen in the GRMTS system. The system is strictly separated by platform: Mobile App (React Native/Expo) for Landlords and Tenants, Web Portal (React) for GRA Officers, Inspectors, and Administrators.

6.1 Platform Overview

6.1.1 Mobile Application (React Native/Expo)

The mobile app is exclusively for Landlords and Tenants. It is built using React Native with Expo platform.

Target Users: Landlords and Tenants only

Technology: React Native 0.73.x, Expo 50.x, React Navigation 6.x

State Management: Zustand 4.x

UI Components: react-native-paper 5.x

6.1.2 Web Portal (React)

The web portal is exclusively for GRA staff, Inspectors, and Administrators. Landlords and Tenants attempting to access the web portal are redirected to download the mobile app.

Target Users: GRA_OFFICER, GRA_SUPERVISOR, INSPECTOR, ADMIN, SYSTEM_ADMIN

Technology: React 18.x, Vite 5.x, TailwindCSS 3.x

State Management: Zustand 4.x

Charts: Recharts 2.x

6.2 Color Theme (Ghana Flag Colors)

The GRMTS color palette is based on the Ghana national flag colors:

Color Name	Hex Code	Usage
Primary (Ghana Green)	#006B3F	Primary buttons, headers, active states
Secondary (Ghana Gold)	#FCD116	Highlights, warnings, accents
Accent (Ghana Red)	#CE1126	Errors, alerts, destructive actions
Background	#F5F5F5	Page backgrounds
Surface	#FFFFFF	Cards, modals, form backgrounds
Text	#1A1A1A	Primary text
Text Light	#999999	Secondary text, placeholders
Success	#4CAF50	Success messages, completed states

Warning	#FF9800	Warning messages
Error	#F44336	Error messages, validation errors

7. Mobile Application Screens

The following screens are implemented in the React Native/Expo mobile application for Landlords and Tenants.

7.1 Authentication Screens

7.1.1 Welcome Screen (MOB-AUTH-001)

Purpose: Initial app entry point with options to login or register

File: src/screens/auth/WelcomeScreen.tsx

Screen Elements:

- GRMTS Logo (centered, prominent)
- App tagline: 'Ghana Rental Market Taxation System'
- 'Login' button (Primary, full width)
- 'Register' button (Secondary, full width)
- Version number at bottom

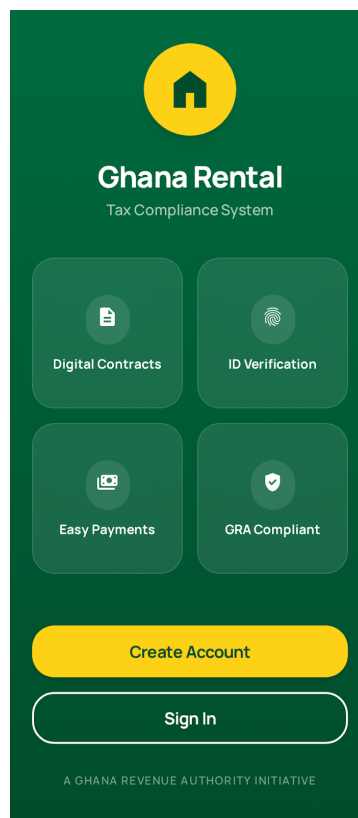


Figure: Welcome Screen - Mobile App Entry Point

7.1.2 Login Screen (MOB-AUTH-002)

Purpose: Authenticate existing users

File: src/screens/auth/LoginScreen.tsx

Field	Required	Type	Validation
Email	Yes	TextInput	Valid email format
Password	Yes	SecureTextEntry	Non-empty

Actions:

- Login button - validates and authenticates
- 'Forgot Password?' link
- 'Don't have an account? Register' link

On Success: Navigate to role-specific dashboard (Landlord or Tenant)

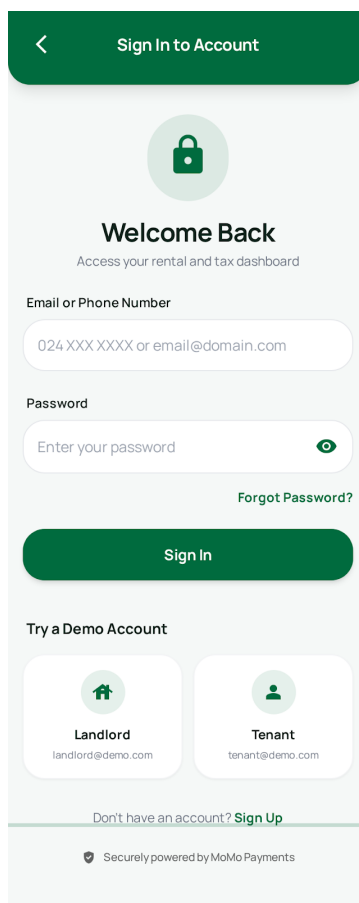


Figure: Login Screen - User Authentication

7.1.3 Registration Screen (MOB-AUTH-003)

Purpose: Create new user account

File: src/screens/auth/RegisterScreen.tsx

Field	Required	Type	Validation
Role	Yes	SegmentedControl	LANDLORD or TENANT

First Name	Yes	TextInput	Min 2 characters
Last Name	Yes	TextInput	Min 2 characters
Email	Yes	TextInput	Valid email, unique
Phone	Yes	TextInput	Ghana format (0XX XXX XXXX)
Ghana Card Number	Yes	TextInput	GHA-XXXXXXXXX-X format
Password	Yes	SecureTextEntry	Min 8 chars, complexity
Confirm Password	Yes	SecureTextEntry	Must match password

On Submit: Create user, send OTP to phone, navigate to verification screen

7.1.4 OTP Verification Screen (MOB-AUTH-004)

Purpose: Verify phone number with OTP code

File: src/screens/auth/VerifyOTPScreen.tsx

Field	Required	Type	Validation
OTP Code	Yes	6-digit input	Matches sent code (demo: 123456)

Display Elements:

- Phone number (partially masked)
- 6 individual input boxes for OTP digits
- Countdown timer for resend (60 seconds)
- 'Resend Code' button (enabled after timer)

On Success: Navigate to dashboard

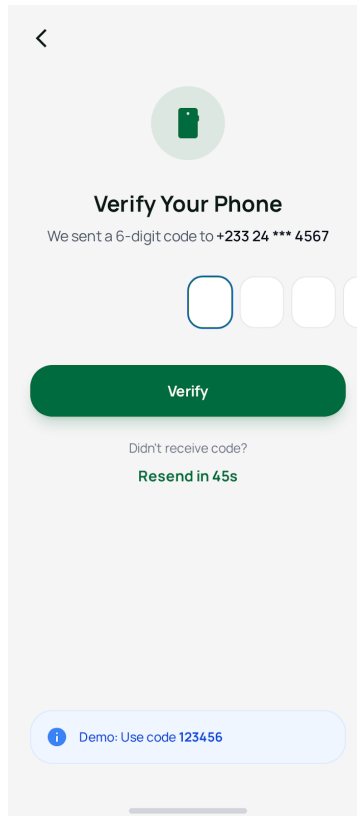


Figure: OTP Verification Screen - Phone Number Verification

7.2 Landlord Screens

7.2.1 Landlord Dashboard (MOB-LND-001)

Purpose: Overview of landlord's properties and contracts

File: src/screens/landlord/DashboardScreen.tsx

Summary Cards:

- Total Properties (count)
- Active Contracts (count)
- Total Revenue (GHS, current month)
- Tax Withheld (GHS, current month)

Quick Actions:

- 'Add Property' button
- 'Create Contract' button
- 'Download Tax Certificate' button

Recent Activity List:

- Recent payments received
- Contract status changes
- New tenant confirmations

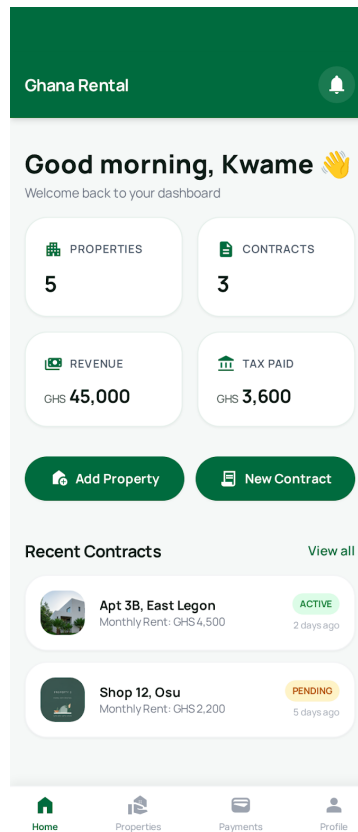


Figure: Landlord Dashboard - Properties and Revenue Overview

7.2.2 Properties List Screen (MOB-LND-002)

Purpose: View and manage registered properties

File: src/screens/landlord/PropertiesScreen.tsx

Screen Elements:

- Search bar (search by address or code)
- Filter chips (All, Verified, Pending, Available, Occupied)
- Property cards with photo, address, status, rent
- Floating '+' button to add new property

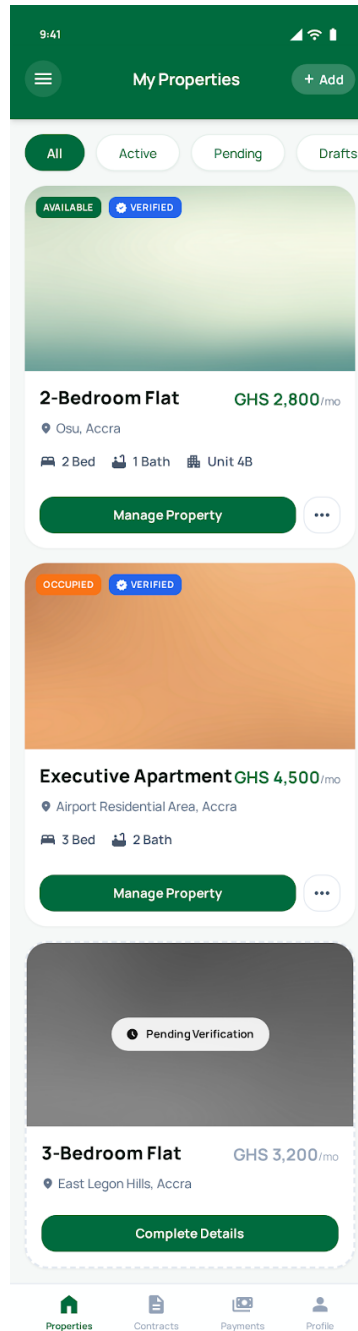


Figure: Properties List Screen - Registered Properties

7.2.3 Add Property Screen (MOB-LND-003)

Purpose: Register a new rental property

File: src/screens/landlord/AddPropertyScreen.tsx

Field	Required	Type	Validation
Digital Address	Yes	TextInput	XX-XXX-XXXX format

Region	Yes	Picker	Select from 16 regions
District	Yes	Picker	Filtered by region
Property Type	Yes	Picker	Apartment, House, etc.
Category	Yes	SegmentedControl	RESIDENTIAL or COMMERCIAL
Bedrooms	If residential	Stepper	0-10+
Bathrooms	If residential	Stepper	0-10+
Is Furnished	No	Switch	Yes/No
Photos	Yes	ImagePicker	Min 1, max 10 photos

Add Property

PROPERTY LOCATION Step 1 of 3

Digital Address Ghana Post GPS

GA-123-4567

Region

Select Region

District

Select District

City / Town

e.g. East Legon

Street Address & Landmarks

House Number, Street name, or closest landmark

Continue to Details

Figure: Add Property Screen - Property Registration Form

7.2.4 Create Contract Screen (MOB-LND-004)

Purpose: Create a new rental contract with ID scanning

File: src/screens/landlord/CreateContractScreen.tsx

Multi-step wizard with progress indicator:

Step 1: Select Property

- Picker to select from verified, available properties

- Shows property photo and address when selected

Step 2: Scan Tenant ID

- Button to open camera for Ghana Card scanning
- Uses expo-camera for image capture
- OCR extracts: Ghana Card Number, Name, Date of Birth, Gender
- Displays extracted data with confidence score
- Option to manually correct extracted data

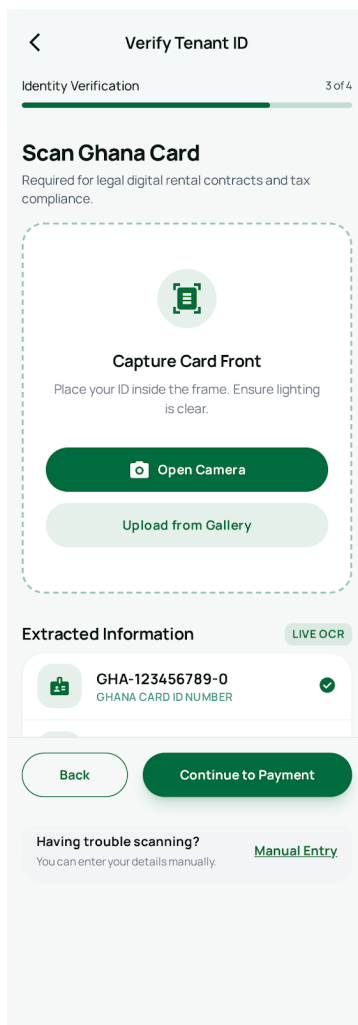


Figure: Scan Tenant ID - Ghana Card Verification

< Verify ID

CAPTURED DOCUMENT



Extraction Confidence 85%

Extracted Information
Please ensure all details match your physical ID card.

GHANA CARD NUMBER
GHA-123456789-0

FULL NAME
Kwame Asante Mensah

DATE OF BIRTH
15 May 1985

GENDER
Male

[Retake](#) [Confirm Details](#)

Figure: Verify Extracted Data - OCR Results Review

Step 3: Contract Terms

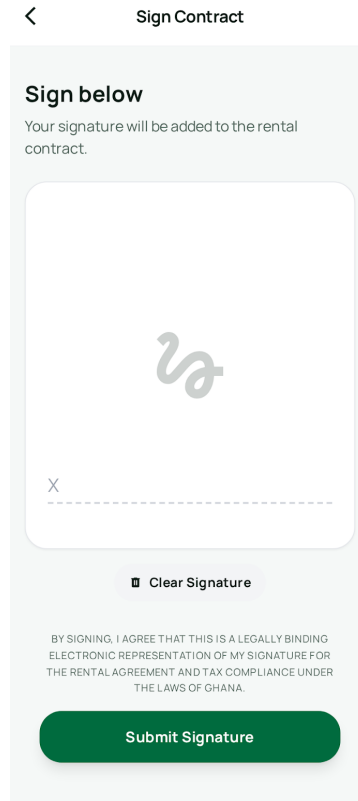
Field	Required	Type	Validation
Start Date	Yes	DatePicker	Today or future
End Date	Yes	DatePicker	After start date
Monthly Rent (GHS)	Yes	NumericInput	Min GHS 50
Advance Months	Yes	Stepper	1-6 residential, 1-24 commercial
Security Deposit	No	NumericInput	Max 3x monthly rent
Service Charge	No	NumericInput	Any positive amount

Live Tax Calculation Display:

- Gross Amount: GHS X,XXX
- Tax (8%): GHS XXX
- Net to Landlord: GHS X,XXX

Step 4: Sign Contract

- Full-screen signature pad (react-native-signature-canvas)
- Clear and Redo buttons
- Signature saved as base64 image
- 'I agree to the terms' checkbox



The image shows a mobile app screen titled "Sign Contract". At the top left is a back arrow. Below the title, the text "Sign below" is followed by "Your signature will be added to the rental contract." A large white rounded rectangle contains a grey stylized signature and a dashed line with an "X" at the end. Below this is a "Clear Signature" button with a trash icon. A legal disclaimer in small text reads: "BY SIGNING, I AGREE THAT THIS IS A LEGALLY BINDING ELECTRONIC REPRESENTATION OF MY SIGNATURE FOR THE RENTAL AGREEMENT AND TAX COMPLIANCE UNDER THE LAWS OF GHANA." At the bottom is a green "Submit Signature" button.

Figure: Digital Signature Pad - Contract Signing

Step 5: Confirmation

- Summary of all contract details
- Display 10-character confirmation code
- Instructions to share code with tenant
- 'Share Code' button (opens device share sheet)
- Code expires in 24 hours

7.3 Tenant Screens

7.3.1 Tenant Dashboard (MOB-TNT-001)

Purpose: Overview of tenant's contracts and payments

File: src/screens/tenant/DashboardScreen.tsx

Current Rental Card:

- Property photo and address
- Landlord name
- Monthly rent amount
- Next payment due date

- 'Pay Now' button

Pending Actions Alert:

- Contracts awaiting confirmation (prominent banner)
- 'View Pending Contract' button

Recent Payments List:

- Last 5 payments with amount, date, status

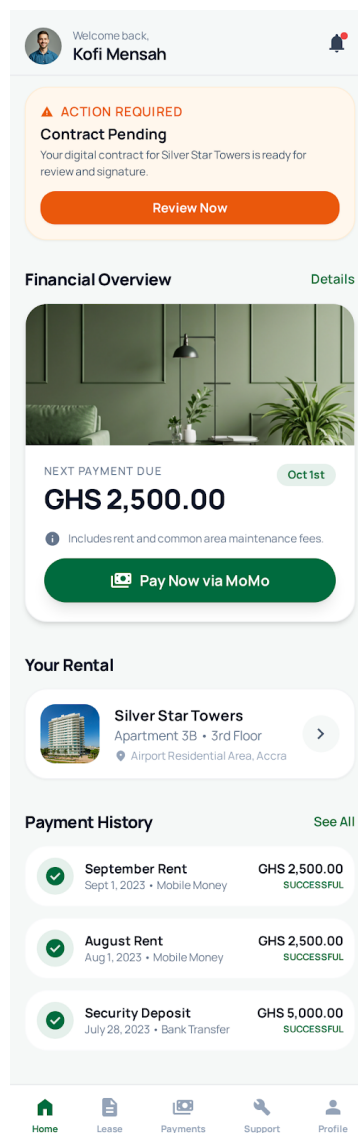


Figure: Tenant Dashboard - Contract and Payment Overview

7.3.2 Confirm Contract Screen (MOB-TNT-002)

Purpose: Review and confirm pending contract

File: src/screens/tenant/ConfirmContractScreen.tsx

Screen Flow:

Step 1: Enter Confirmation Code

Field	Required	Type	Validation
Confirmation Code	Yes	TextInput	10-character code from landlord

On Submit: Fetch contract details if code is valid

Step 2: Review Contract Details

Display:

- Property information (address, type, features)
- Landlord information (name, contact)
- Contract terms (duration, rent, advance, deposit)
- Payment breakdown with 8% tax display

Actions: 'Accept' button, 'Decline' button

Step 3: Scan Own ID

- Camera opens to scan tenant's own Ghana Card
- OCR extracts and verifies identity
- Confirms Ghana Card matches registered user

Step 4: Sign Contract

- Full-screen signature pad
- Signature saved as base64 image
- 'I confirm this contract' checkbox

Step 5: Confirmation Success

- Success animation
- Contract now ACTIVE
- 'Make First Payment' button

7.4 Shared Mobile Screens

7.4.1 Document Scan Screen (MOB-SHR-001)

Purpose: Scan Ghana Card using device camera with OCR

File: src/screens/shared/ScanDocumentScreen.tsx

Screen Elements:

- Full-screen camera view (expo-camera)
- Rectangular overlay guide for card positioning
- 'Capture' button (centered bottom)
- 'Use Gallery' button (bottom left)
- Flash toggle (top right)

OCR Processing:

- Loading spinner during OCR processing
- Extracts Ghana Card data fields
- Displays confidence score (0-100%)

Extracted Data Interface:

Field	Format
ghanaCardNumber	GHA-XXXXXXXXX-X format
fullName	Complete name as on card
firstName	Parsed first name
lastName	Parsed last name
dateOfBirth	YYYY-MM-DD format
gender	M or F
nationality	e.g., GHANAIAN
confidence	0-100 OCR confidence score

7.4.2 Digital Signature Screen (MOB-SHR-002)

Purpose: Capture digital signature for contracts

File: src/screens/shared/SignContractScreen.tsx

Screen Elements:

- Full-screen white signature canvas (react-native-signature-canvas)
- 'Clear' button (top left) - erases signature
- 'Done' button (top right) - saves signature
- Instructions text: 'Sign within the box below'

Signature Output:

- Base64-encoded PNG image
- Uploaded to server for storage
- Linked to contract record

7.4.3 Make Payment Screen (MOB-SHR-003)

Purpose: Process rent payment via mobile money

File: src/screens/shared/MakePaymentScreen.tsx

Payment Provider Selection:

- MTN Mobile Money (with logo)
- Vodafone Cash (with logo)
- AirtelTigo Money (with logo)

Field	Required	Type	Validation
Phone Number	Yes	TextInput	Ghana mobile format

Payment Breakdown Display:

- Gross Amount: GHS X,XXX.XX
- Withholding Tax (8%): GHS XXX.XX
- Net to Landlord: GHS X,XXX.XX

Payment Flow:

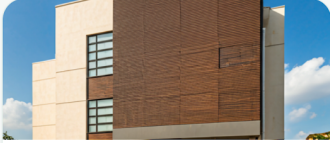
1. User selects provider and enters phone number
2. User taps 'Pay Now' button
3. Loading screen: 'Awaiting payment approval'
4. User receives prompt on mobile phone to authorize
5. User enters PIN on their phone (external)
6. Success screen with receipt details

Payment Status Values:

- PENDING - Payment initiated, awaiting authorization
- PROCESSING - Authorization received, processing
- COMPLETED - Payment successful
- FAILED - Payment failed (with reason)
- REFUNDED - Payment was refunded

<

Make Payment



RENT SUMMARY

FEB 2025

GHS 2,700.00

Monthly Rent	GHS 2,500.00
Income Tax (8%)	GHS 200.00
Net to Landlord	GHS 2,500.00

Select Payment Method

MTN

MTN Mobile Money

Fast & Reliable

✓

Vodafone Cash

Instant transfer

AirtelTigo Money

Pay GHS 2,700.00 →

SECURE RENTAL TRANSACTION

+233 024 456 7890

Secured by Bank of Ghana compliant payment gateway

Figure: Make Payment Screen - Mobile Money Payment

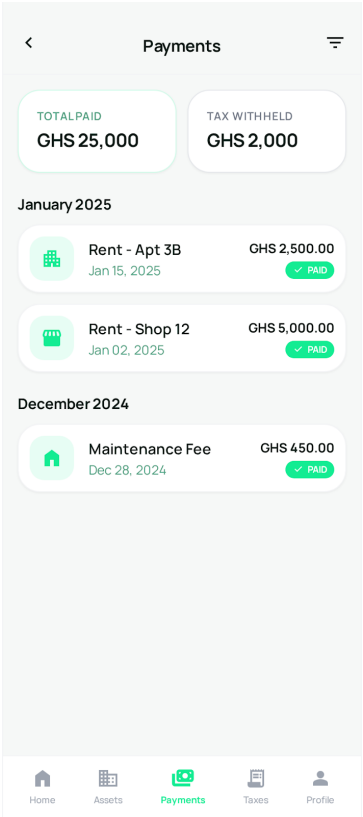


Figure: Payment History - Transaction Records

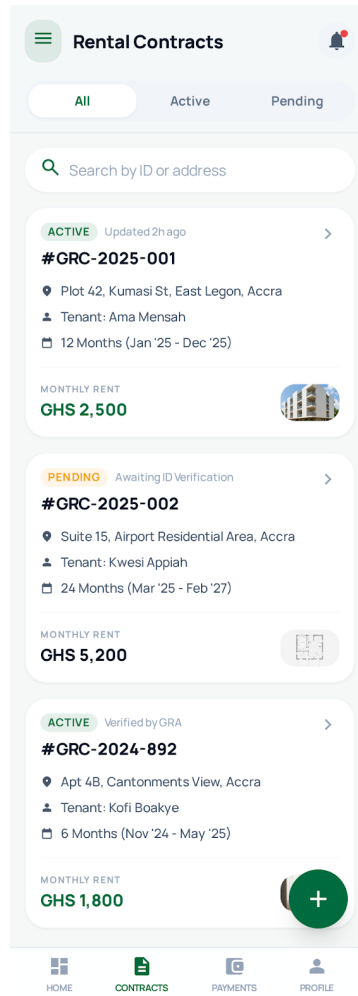


Figure: Rental Contracts List - Contract Management

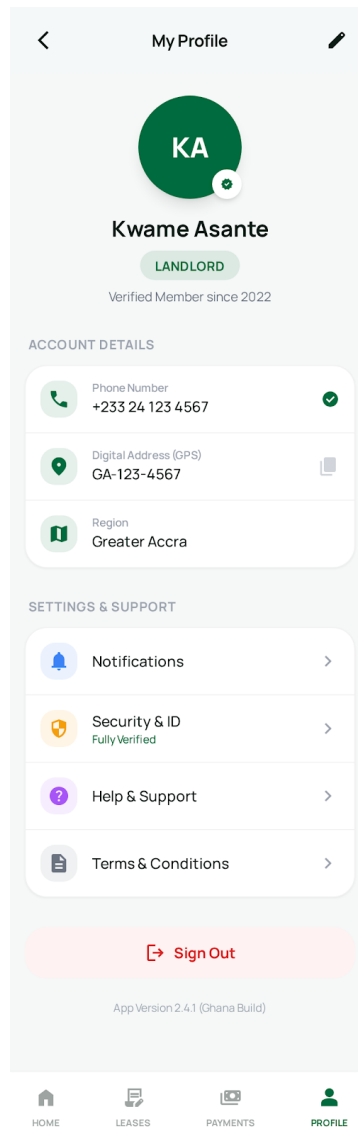


Figure: My Profile - User Profile Management

8. Web Portal Screens

The following screens are implemented in the React web portal for GRA Officers, Inspectors, and Administrators.

8.1 Access Restriction

When Landlords or Tenants attempt to login via the web portal, they receive a redirect message:

- Title: 'Mobile App Required'
- Message: 'Landlords and Tenants must use the GRMTS Mobile App'
- App Store and Play Store download links
- QR code for quick download

Ghana Rental Authority Portal

Email Address
e.g. officer@gra.gov.gh

Password
Enter your password

☐ Remember me [Forgot password?](#)

Sign In

DEMO ACCOUNT ACCESS

- GRA Officer
- Inspector
- Administrator

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Figure: GRA Portal Login - Web Portal Access

8.2 GRA Officer Screens

8.2.1 GRA Dashboard (WEB-GRA-001)

Purpose: Overview of tax collection and compliance metrics

URL: /dashboard

KPI Cards:

- Total Tax Collected (this month) - GHS amount
- Active Contracts - count
- Registered Landlords - count
- Compliance Rate - percentage

Charts:

- Tax Collection Trend (Recharts line chart, 12 months)
- Contracts by Region (Recharts bar chart)
- Payment Method Distribution (Recharts pie chart)

Recent Activity Table:

- Recent payments with contract number, amount, tax withheld
- Sortable columns, pagination

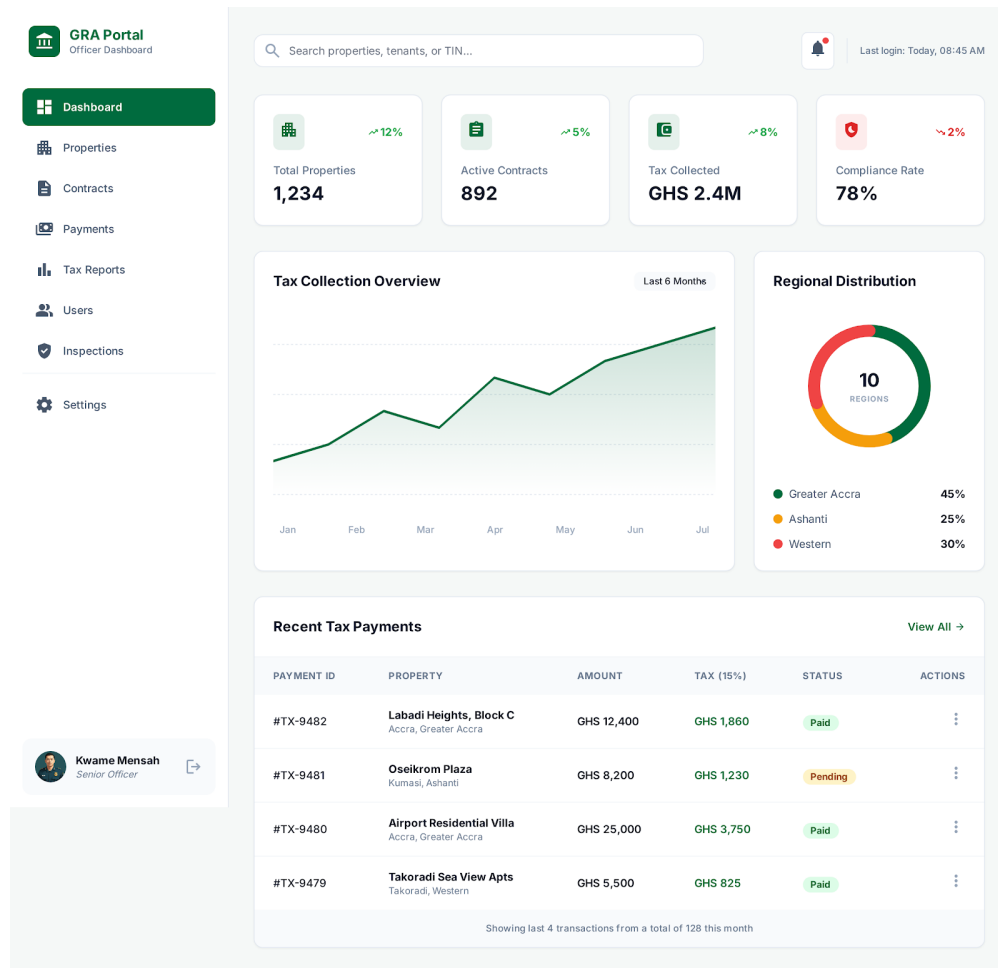


Figure: GRA Officer Dashboard - Tax Collection Overview

8.2.2 Contracts List (WEB-GRA-002)

Purpose: Search and view all registered contracts

URL: /contracts

Filters:

- Status: All, Active, Pending, Expired, Terminated
- Region: Dropdown of 16 regions
- Date Range: Start/End date pickers
- Search: Contract number, landlord name, property address

Table Columns:

- Contract Number, Landlord Name, Tenant Name, Property Address
- Monthly Rent, Start Date, End Date, Status
- Actions: View Details

Export Options:

- Export to CSV button
- Export to PDF button

The screenshot displays the GRA Portal's Property Management interface. The top navigation bar includes 'Dashboard', 'Properties' (active), 'Inspections', 'Reports', and 'Users'. The main section is titled 'Property Management' with a subtitle 'Manage and verify property listings across Ghana's regions.' and an 'Add New Property' button. Below this is a filter bar with tabs for 'All', 'Verified', 'Pending', and 'Flagged'. A search bar and dropdowns for 'Region: Greater Accra' and 'Status: All' are present. A table lists three properties:

PROPERTY	OWNER	REGION	STATUS	ACTIONS
☒ Vista Heights Apartments GA-123-4567, Airport Residential Area	Kwame Mensah GH-ID-2023-9981	Greater Accra	Verified	⋮
☐ Sunrise Villa EK-002-9811, Kumasi	Abena Osei GH-ID-2021-0042	Ashanti	Pending	⋮
☐ Central Plaza Offices AD-441-2090, Adabraka	John Dumelo Ltd GH-TAX-990112	Greater Accra	Flagged	⋮

The right sidebar provides details for 'Vista Heights Apartments' (ID: ACC-20301-P). It includes a location photo, address '124 Independence Avenue, Airport Residential Area, Accra', digital address 'GA-123-4567', and map coordinates. Under 'OWNER CONTACTS', it lists 'Kwame Mensah' (Verified Individual Owner) with phone '+233 24 567 8910', email 'k.mensah@ghanalink.com', and ID 'GHA-722100431-2'. The 'FINANCIAL SUMMARY' shows 'MONTHLY RENT: ₵4,500' and 'RENT TAX (8%): ₵360'. At the bottom are buttons for 'Request Inspection' and 'Flag Property'.

Figure: Property Management - Contract and Property Details

8.2.3 Tax Reports (WEB-GRA-003)

Purpose: Generate and view tax collection reports**URL:** /reports/tax**Report Types:**

- Daily Collection Summary
- Monthly Collection by Region
- Landlord Tax Withholding Details
- Year-to-Date Summary

Report Parameters:

- Period: Date range selection
- Region: Optional filter

- Format: View on screen or download PDF/CSV

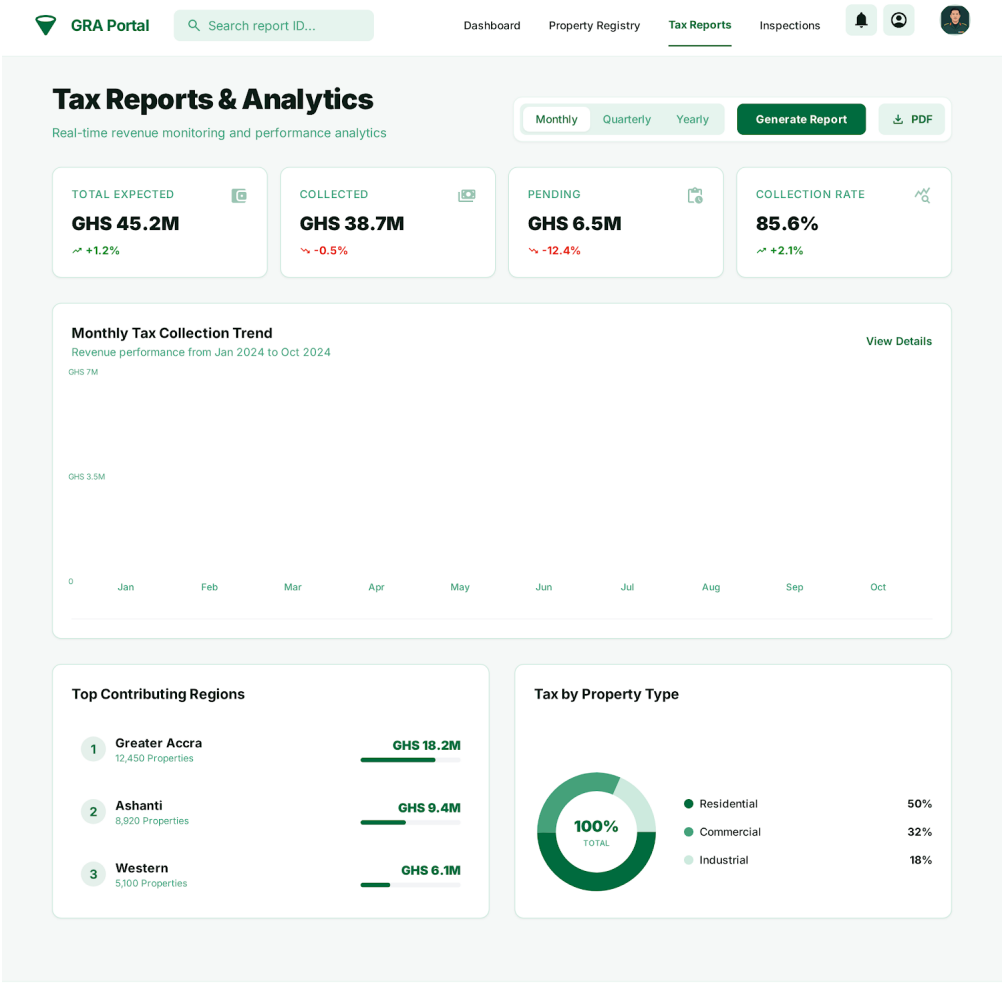


Figure: Tax Reports & Analytics - Revenue Analysis

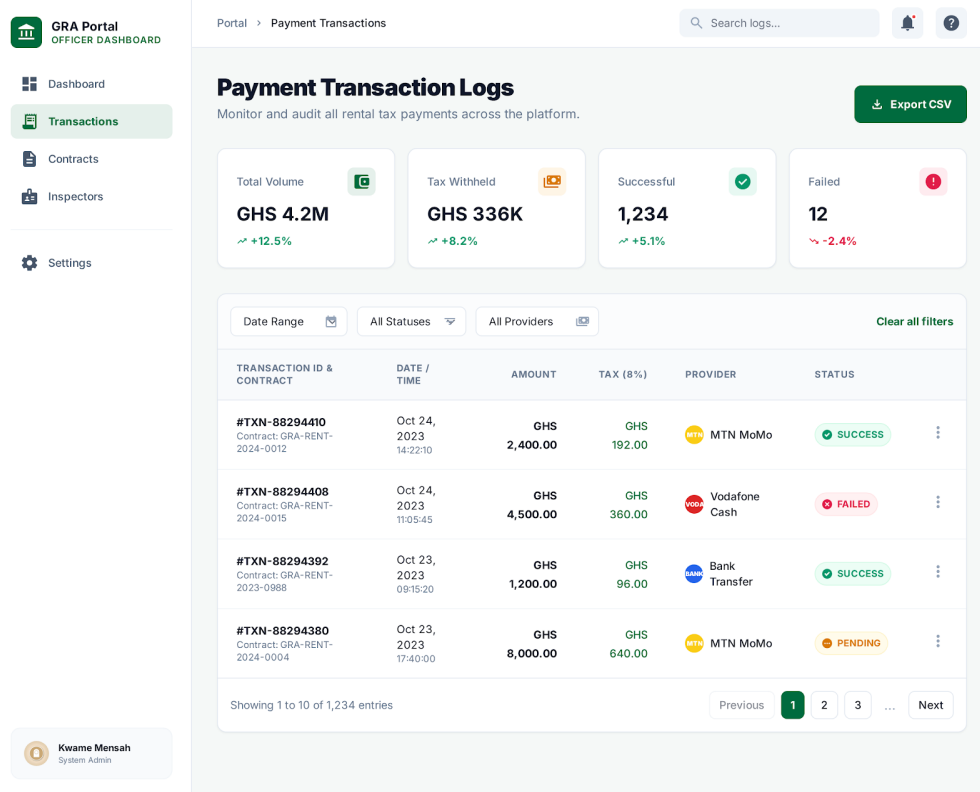


Figure: Payment Transaction Logs - Payment Records

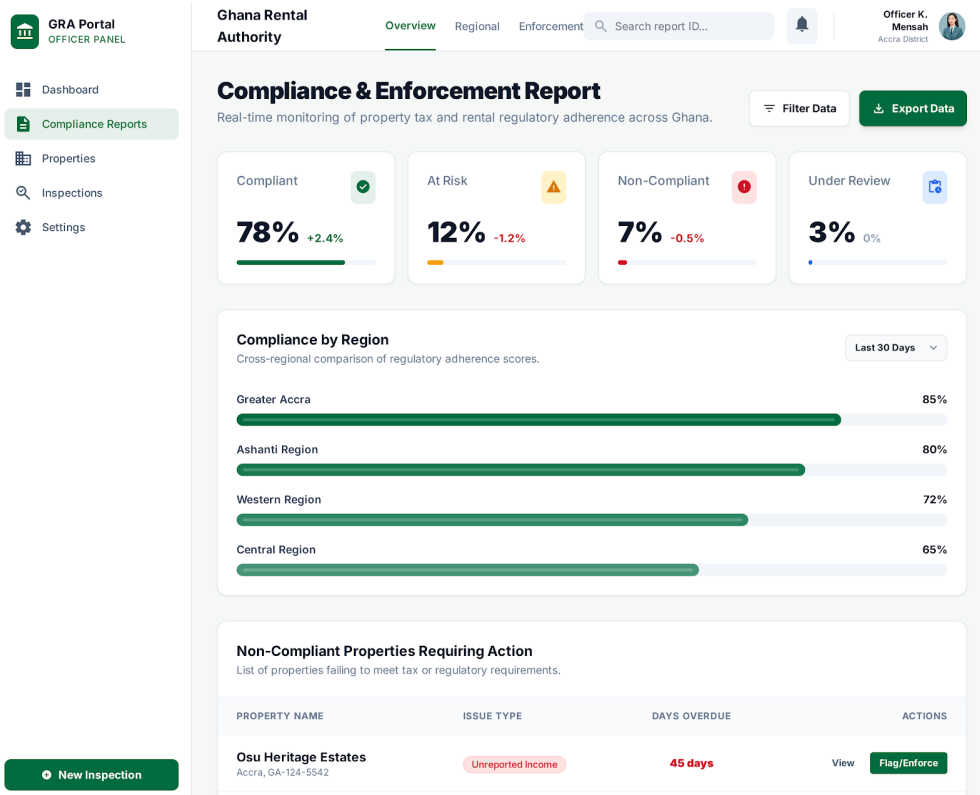


Figure: Compliance & Enforcement Report - Compliance Monitoring

8.3 Inspector Screens

8.3.1 Inspector Dashboard (WEB-INS-001)

Purpose: View and manage assigned inspection cases

URL: /inspections

Assigned Cases:

- List of cases assigned to logged-in inspector
- Case status: Pending, In Progress, Completed
- Due date, priority level

Case Actions:

- View Case Details
- Update Status
- Upload Evidence
- Submit Report

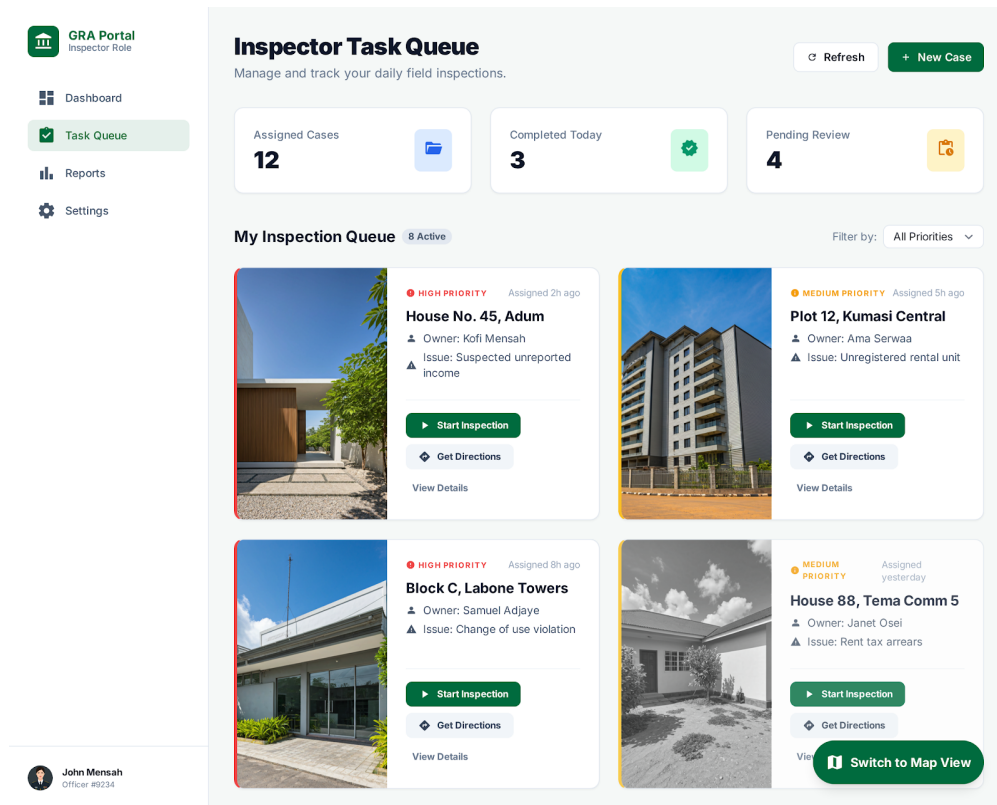


Figure: Inspector Task Queue - Assigned Cases

8.3.2 Case Details (WEB-INS-002)

Purpose: View and update specific inspection case

URL: /inspections/:id

Case Information:

- Property address and details
- Landlord and tenant information
- Contract details if applicable
- Case notes and history

Evidence Upload:

- Photo upload (drag-and-drop)
- Document upload (PDF, images)
- Evidence description field

Report Submission:

- Findings text area
- Recommendation dropdown
- Submit Report button

8.4 Admin Screens

8.4.1 User Management (WEB-ADM-001)

Purpose: Manage system user accounts

URL: /admin/users

User List:

- Search by name, email, phone
- Filter by role, status
- Table: Name, Email, Role, Status, Created Date

User Actions:

- View User Details
- Edit User (change role, status)
- Deactivate User
- Reset Password

Add New User
Register a new official onto the GRA portal

FIRST NAME
e.g. Samuel

LAST NAME
e.g. Mensah

EMAIL ADDRESS
samuel.mensah@gra.gov.gh

PHONE NUMBER
+233 24 XXX XXXX

SYSTEM ROLE
Select Role

ASSIGNED REGION
Select Region

Cancel Create User

Figure: Admin User Management - User Administration

8.4.2 System Configuration (WEB-ADM-002)

Purpose: Configure system settings

URL: /admin/settings

Configuration Sections:

- Tax Settings (rate display - read only in current implementation)
- Notification Templates
- Reference Data (regions, property types)
- API Keys Management
- System Health Monitoring

GRA Admin Portal

DashboardUsersInspectionsReportsSettings

Search system logs...

System Settings & Integrations

Global portal configuration for Ghana Rental Authority.

Audit Logs

GeneralTax ConfigNotificationsIntegrationsSecurity

Tax Configuration

Withholding Tax Rate (%)

8.0%

Minimum Taxable Rent (GHS)

GHS 0

Tax Payment Due Day

15th

Mobile Money Integration

MTN

CONNECTED

MTN Mobile Money

Primary settlement account for all rent taxes.

Configure

VOD

DISCONNECTED

Vodafone Cash

Enable payments via Vodafone Ghana subscribers.

Setup

ATO

DISCONNECTED

AirtelTigo Money

Allow rent tax payments via AirtelTigo wallet.

Setup

Last updated by Admin-04 on Oct 24, 2023 at 14:32

Cancel

Save Changes

Figure: System Settings & Integrations - Configuration Panel

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7. Business Rules

This section documents all business rules that govern system behavior. Rules are organized by functional area and include examples to illustrate application.

7.1 Contract Business Rules

7.1.1 Advance Payment Limits

Rule ID	Rule Description	Property Type	Limit
BR-CTR-001	Maximum advance for single room	R-SR	6 months
BR-CTR-002	Maximum advance for self-contained	R-SC	6 months
BR-CTR-003	Maximum advance for apartments (1-4 bed)	R-1B to R-4B	6 months
BR-CTR-004	Maximum advance for houses/villas	R-HS, R-VL	6 months
BR-CTR-005	Maximum advance for shops	C-SHP	12 months
BR-CTR-006	Maximum advance for offices	C-OFF	12 months
BR-CTR-007	Maximum advance for warehouses	C-WHS	24 months
BR-CTR-008	Maximum advance for industrial	C-IND	24 months
BR-CTR-009	Maximum advance for mixed use	C-MXD	12 months

Example Application:

Scenario: Landlord attempts to create a contract for a 2-bedroom apartment (R-2B) with 12 months advance.

Rule Applied: BR-CTR-003 limits R-2B to 6 months maximum.

System Response: Error message 'Advance payment of 12 months exceeds the maximum allowed (6 months) for this property type. Please reduce the advance period.'

7.1.2 Security Deposit Rules

Rule ID	Rule Description	Value
BR-CTR-010	Maximum security deposit as multiple of monthly rent	3 months
BR-CTR-011	Security deposit must be refundable under agreed terms	Required
BR-CTR-012	Security deposit is not subject to withholding tax	0% tax

Example Calculation:

Monthly Rent: GHS 2,000

Maximum Security Deposit: $\text{GHS } 2,000 \times 3 = \text{GHS } 6,000$

If landlord enters GHS 8,000, system displays: 'Security deposit of GHS 8,000 exceeds the maximum allowed (GHS 6,000 = 3 months rent).'

7.1.3 Contract Duration Rules

Rule ID	Rule Description	Value
BR-CTR-020	Minimum contract duration	1 month
BR-CTR-021	Maximum contract duration	60 months (5 years)
BR-CTR-022	Contract start date cannot be more than X days in the past	90 days
BR-CTR-023	Tenant confirmation period	7 days
BR-CTR-024	Minimum termination notice period	30 days

7.1.4 Contract Validation Rules

Rule ID	Rule Description
BR-CTR-030	Contract cannot be created for unverified properties
BR-CTR-031	Contract cannot be created for properties with existing active contracts (unless multi-unit)
BR-CTR-032	Landlord cannot create contract with themselves as tenant
BR-CTR-033	Contract end date must be after start date
BR-CTR-034	Advance months cannot exceed contract duration
BR-CTR-035	Monthly rent must be positive and at least GHS 50
BR-CTR-036	Contract document must be uploaded before submission

7.2 Tax Calculation Rules**7.2.1 Withholding Tax Rate**

GRMTS applies a flat 8% withholding tax on all rent payments processed through the system.

Rule ID	Rule Description	Value
BR-TAX-001	Standard withholding tax rate on all rent payments	8%
BR-TAX-002	Tax is automatically calculated: $\text{tax_amount} = \text{gross_amount} \times 0.08$	Automatic
BR-TAX-003	Tax is deducted at source before landlord receives funds	At source
BR-TAX-004	Tax rate is consistent regardless of landlord type or TIN status	Universal
BR-TAX-005	Tax is remitted directly to Ghana Revenue Authority	Direct remittance

7.2.2 Tax Certificate Rules

Rule ID	Rule Description	Value
BR-TAX-010	Landlords can download tax certificates from mobile app	Available
BR-TAX-011	Certificates show total tax withheld for filing purposes	PDF download
BR-TAX-012	Certificates are generated per contract period	Per contract

Tax Calculation Example:

Payment Amount: GHS 1,000 (monthly rent)

Calculation:

- Gross Amount: GHS 1,000.00
- Tax Rate: 8% (BR-TAX-001)
- Tax Amount: $\text{GHS } 1,000 \times 0.08 = \text{GHS } 80.00$
- Net to Landlord: $\text{GHS } 1,000 - \text{GHS } 80 = \text{GHS } 920.00$

Tax Display During Payment:

The payment screen clearly shows the breakdown:

- Gross Amount: GHS 1,000.00
- Tax (8%): GHS 80.00
- Net to Landlord: GHS 920.00

7.3 Payment Rules

7.3.1 Mobile Money Payment Rules

Rule ID	Rule Description	Value
BR-PAY-001	Supported payment providers	MTN, Vodafone, AirtelTigo
BR-PAY-002	User must enter mobile money phone number	Required
BR-PAY-003	User authorizes payment on their mobile phone	External PIN entry
BR-PAY-004	Payment confirmation received via callback	Webhook/polling

Payment Flow:

1. Tenant selects payment provider (MTN, Vodafone, or AirtelTigo)
2. Tenant enters mobile money phone number
3. Tenant taps 'Pay Now' button
4. System initiates payment request to provider

5. User receives prompt on their mobile phone
6. User enters PIN to authorize payment
7. Payment confirmation is received by the system
8. Receipt is generated and displayed

7.3.2 Payment Validation Rules

Rule ID	Rule Description
BR-PAY-010	Payments can only be made for active contracts
BR-PAY-011	Minimum payment amount is one month's rent
BR-PAY-012	Duplicate payments for same period are blocked within 5 minutes
BR-PAY-013	Payment timeout after 5 minutes of processing
BR-PAY-014	Failed payments may be retried up to 3 times
BR-PAY-015	Settlement batches run at 6:00 AM and 6:00 PM daily

7.4 Compliance and Enforcement Rules

7.4.1 Compliance Score Calculation

Rule ID	Factor	Weight	Max Impact
BR-CMP-001	Contract registration rate (% of properties with registered contracts)	30%	-30 points
BR-CMP-002	Payment timeliness (% of payments received on/before due date)	25%	-25 points
BR-CMP-003	Tenant confirmation rate (% of contracts confirmed within 7 days)	15%	-15 points
BR-CMP-004	Document completeness (ownership verified)	10%	-10 points
BR-CMP-005	Dispute history (deduction per lost dispute)	10%	-10 points
BR-CMP-006	Inspection results (deduction per confirmed violation)	10%	-10 points

Compliance Score Example:

Landlord Profile:

- 5 properties, 4 with registered contracts (80%) → Score impact: -6 points (30% × 20%)
- 90% payments on time → Score impact: -2.5 points (25% × 10%)
- 100% tenant confirmations → Score impact: 0 points
- All ownership docs verified → Score impact: 0 points
- 1 lost dispute → Score impact: -5 points
- No violations → Score impact: 0 points

Total Score: 100 - 6 - 2.5 - 0 - 0 - 5 - 0 = 86.5 (rounded to 87)

7.4.2 Risk Score Calculation

Rule ID	Risk Factor	Points
BR-RSK-001	Property registered but no contracts for >6 months	+25
BR-RSK-002	GSM presence detected but no registered contract	+20
BR-RSK-003	Utility consumption anomaly (high usage, no contract)	+15
BR-RSK-004	Anonymous tip received (per unique tip)	+15
BR-RSK-005	Previous violations by landlord (per violation)	+10
BR-RSK-006	Contract irregularities detected	+10
BR-RSK-007	Threshold for automatic case creation	>60 points

8. Data Dictionary

This section defines all data entities, attributes, and relationships in the GRMTS database.

8.1 User Entity

Field Name	Data Type	Required	Description
id	UUID	Yes	Unique identifier (system generated)
email	VARCHAR(255)	Yes	Email address, unique constraint
phone	VARCHAR(20)	Yes	Phone number, unique constraint
password_hash	VARCHAR(255)	Yes	Bcrypt hashed password
role	ENUM	Yes	User role (LANDLORD_INDIVIDUAL, etc.)
first_name	VARCHAR(100)	Yes	First name from Ghana Card
last_name	VARCHAR(100)	Yes	Last name from Ghana Card
ghana_card_number	VARCHAR(20)	Yes	Ghana Card ID, unique, format GHA-XXXXXXXX-X
tin_number	VARCHAR(20)	No	Taxpayer Identification Number
digital_address	VARCHAR(20)	No	Ghana Post digital address
region	VARCHAR(50)	No	Region of residence
district	VARCHAR(50)	No	District of residence
is_corporate	BOOLEAN	Yes	True if corporate account
company_name	VARCHAR(255)	If corporate	Registered company name
status	ENUM	Yes	PENDING, ACTIVE, SUSPENDED, BLACKLISTED
compliance_score	INTEGER	Yes	0-100 compliance score (landlords)
created_at	TIMESTAMP	Yes	Account creation timestamp
updated_at	TIMESTAMP	Yes	Last update timestamp

8.2 Property Entity

Field Name	Data Type	Required	Description
id	UUID	Yes	Unique identifier
landlord_id	UUID (FK)	Yes	Reference to users table
property_code	VARCHAR(30)	Yes	Unique property code, system generated
digital_address	VARCHAR(20)	Yes	Ghana Post GPS address
region	VARCHAR(50)	Yes	Region (from digital address)
district	VARCHAR(50)	Yes	District (from digital address)
property_type	ENUM	Yes	R-SR, R-SC, R-1B, etc.
property_category	ENUM	Yes	RESIDENTIAL or COMMERCIAL
bedrooms	INTEGER	If residential	Number of bedrooms (0-10+)
bathrooms	INTEGER	No	Number of bathrooms

floor_area_sqm	DECIMAL(10,2)	No	Floor area in square meters
is_furnished	ENUM	Yes	UNFURNISHED, SEMI, FULLY
amenities	JSONB	No	Array of amenity codes
ownership_type	ENUM	Yes	FREEHOLD, LEASEHOLD, etc.
ownership_doc_url	TEXT	Yes	S3 URL to ownership document
photos	JSONB	Yes	Array of photo URLs
status	ENUM	Yes	PENDING, VERIFIED, SUSPENDED
is_available	BOOLEAN	Yes	Available for new contracts
created_at	TIMESTAMP	Yes	Registration timestamp

8.3 Contract Entity

Field Name	Data Type	Required	Description
id	UUID	Yes	Unique identifier
contract_number	VARCHAR(30)	Yes	Human-readable contract number
property_id	UUID (FK)	Yes	Reference to properties table
landlord_id	UUID (FK)	Yes	Reference to users table
tenant_id	UUID (FK)	Yes	Reference to users table
start_date	DATE	Yes	Contract start date
end_date	DATE	Yes	Contract end date
monthly_rent	DECIMAL(12,2)	Yes	Monthly rent in GHS
security_deposit	DECIMAL(12,2)	No	Security deposit in GHS
advance_months	INTEGER	Yes	Number of advance months
tax_rate	DECIMAL(5,4)	Yes	Applicable tax rate (0.08, 0.15, etc.)
total_tax_withheld	DECIMAL(12,2)	Yes	Running total of tax withheld
status	ENUM	Yes	DRAFT, PENDING, ACTIVE, EXPIRED, etc.
confirmation_code	VARCHAR(10)	Yes	6-digit OTP for tenant confirmation
tenant_confirmed_at	TIMESTAMP	No	When tenant confirmed
created_at	TIMESTAMP	Yes	Contract creation timestamp

8.4 Payment Entity

Field Name	Data Type	Required	Description
id	UUID	Yes	Unique identifier
payment_reference	VARCHAR(30)	Yes	Human-readable payment reference
contract_id	UUID (FK)	Yes	Reference to contracts table
tenant_id	UUID (FK)	Yes	Paying tenant
landlord_id	UUID (FK)	Yes	Receiving landlord
gross_amount	DECIMAL(12,2)	Yes	Total payment before deductions

tax_amount	DECIMAL(12,2)	Yes	Tax withheld
platform_fee	DECIMAL(12,2)	Yes	Platform fee charged
net_amount	DECIMAL(12,2)	Yes	Amount to landlord
period_start	DATE	Yes	Start of covered period
period_end	DATE	Yes	End of covered period
payment_method	ENUM	Yes	MTN_MOMO, VODAFONE_CASH, etc.
provider_reference	VARCHAR(100)	No	Payment provider transaction ID
status	ENUM	Yes	PENDING, PROCESSING, COMPLETED, FAILED
initiated_at	TIMESTAMP	Yes	Payment initiation time
completed_at	TIMESTAMP	No	Payment completion time

9. Testing and Quality Assurance

This section defines the testing approach, test cases, and acceptance criteria for the GRMTS project.

9.1 Testing Strategy

The GRMTS will undergo comprehensive testing at multiple levels to ensure quality and reliability.

9.1.1 Testing Levels

Level	Scope	Responsibility
Unit Testing	Individual functions and methods	Development team
Integration Testing	API endpoints, database interactions	Development team
System Testing	End-to-end workflows	QA team
UAT	Business scenarios, user acceptance	Business stakeholders
Performance Testing	Load, stress, scalability	QA team
Security Testing	Vulnerability assessment, penetration testing	Security team

9.2 Test Cases

9.2.1 User Registration Test Cases

TC ID	Test Case	Expected Result	Priority
TC-REG-001	Register with valid email, phone, password	OTP sent, proceed to verification	Critical
TC-REG-002	Register with existing email	Error: Email already registered	Critical
TC-REG-003	Register with existing phone	Error: Phone already registered	Critical
TC-REG-004	Register with invalid email format	Error: Invalid email format	High
TC-REG-005	Register with invalid phone format	Error: Invalid phone number	High
TC-REG-006	Register with weak password	Error: Password does not meet requirements	High
TC-REG-007	Verify phone with correct OTP	Proceed to role selection	Critical
TC-REG-008	Verify phone with incorrect OTP (3 times)	Error: Invalid code, locked	High
TC-REG-009	Verify phone with expired OTP	Error: Code expired, resend	High
TC-REG-010	Verify Ghana Card with valid number	NIA data displayed for confirmation	Critical
TC-REG-011	Verify Ghana Card already in use	Error: Ghana Card registered to another account	Critical
TC-REG-012	Complete landlord registration	Account created, flat 8% tax rate applies	Critical
TC-REG-013	Complete tenant registration	Account created, can confirm contracts	Critical

9.2.2 Contract Management Test Cases

TC ID	Test Case	Expected Result	Priority
TC-CTR-001	Create contract with valid terms	Contract created, SMS sent to tenant	Critical
TC-CTR-002	Create contract with excess advance (residential)	Error: Advance exceeds 6 months	Critical
TC-CTR-003	Create contract with excess deposit	Error: Deposit exceeds 3 months	Critical
TC-CTR-004	Create contract for unverified property	Error: Property not verified	High
TC-CTR-005	Create contract for occupied property	Error: Property has active contract	High
TC-CTR-006	Tenant confirms with valid OTP	Contract status changes to ACTIVE	Critical
TC-CTR-007	Tenant confirms with expired OTP	Error: Confirmation code expired	High
TC-CTR-008	Tenant objects to contract	Status changes to DISPUTED, case created	High
TC-CTR-009	Contract expires without confirmation	Status changes to EXPIRED_UNCONFIRMED	High
TC-CTR-010	Landlord initiates renewal	New contract created linked to original	Medium

9.2.3 Payment Processing Test Cases

TC ID	Test Case	Expected Result	Priority
TC-PAY-001	Make payment via MTN MoMo	Payment completes, receipt generated	Critical
TC-PAY-002	Make payment via Vodafone Cash	Payment completes, receipt generated	Critical
TC-PAY-003	Make payment via AirtelTigo Money	Payment completes, receipt generated	Critical
TC-PAY-004	Verify flat 8% tax calculation	Tax = Gross $\times$ 0.08 for all payments	Critical
TC-PAY-005	Verify net amount to landlord	Net = Gross - Tax (Gross $\times$ 0.92)	Critical
TC-PAY-006	Verify payment breakdown display	Shows Gross, Tax (8%), Net amounts	Critical
TC-PAY-007	Payment fails - insufficient funds	Error message, retry option	High
TC-PAY-008	Payment fails - timeout	Error message, status FAILED	High
TC-PAY-009	Payment status tracking	PENDING -> PROCESSING -> COMPLETED	High
TC-PAY-010	Tax certificate download	Landlord can download certificate from app	Critical

9.3 Requirements Traceability Matrix

The following matrix maps requirements to test cases and acceptance criteria:

Req ID	Requirement	Test Case(s)	Status
USR-FR-001	Registration form with email, phone, password	TC-REG-001 to 006	Pending
USR-FR-005	OTP sent within 30 seconds	TC-REG-001, TC-REG-014	Pending
USR-FR-009	Ghana Card verification with NIA	TC-REG-010, TC-REG-011	Pending
CTR-FR-011	Enforce advance limits by property type	TC-CTR-002	Pending
CTR-FR-016	Enforce 3-month deposit maximum	TC-CTR-003	Pending
CTR-FR-045	Tenant confirmation via OTP	TC-CTR-006, TC-CTR-007	Pending
PAY-FR-008	Flat 8% tax on all payments	TC-PAY-004	Pending
PAY-FR-009	Tax calculation formula	TC-PAY-004, TC-PAY-005	Pending
TAX-FR-028	Tax certificate download	TC-PAY-010	Pending

10. Security Requirements

This section defines the security requirements for protecting GRMTS data and systems.

10.1 Authentication and Access Control

Req ID	Security Requirement
SEC-001	All passwords must be hashed using bcrypt with a cost factor of at least 12
SEC-002	User sessions shall be managed using JWT tokens with 30-minute expiry
SEC-003	Account lockout shall be enforced after 5 failed login attempts for 30 minutes
SEC-004	All sensitive operations shall require OTP verification (two-factor authentication)
SEC-005	Role-based access control shall be enforced for all API endpoints
SEC-006	Password reset tokens shall expire after 15 minutes
SEC-007	All authentication events shall be logged with timestamp, IP, and user agent

10.2 Data Protection

Req ID	Security Requirement
SEC-010	All data in transit shall be encrypted using TLS 1.2 or higher
SEC-011	All sensitive data at rest shall be encrypted using AES-256
SEC-012	Personal data shall be handled in compliance with Ghana Data Protection Act
SEC-013	Payment card data shall not be stored (use tokenization)
SEC-014	Database backups shall be encrypted and stored in secure location
SEC-015	Audit logs shall be tamper-proof and retained for minimum 7 years

10.3 Application Security

Req ID	Security Requirement
SEC-020	All user input shall be validated and sanitized to prevent injection attacks
SEC-021	Cross-Site Scripting (XSS) protection shall be implemented for all outputs
SEC-022	Cross-Site Request Forgery (CSRF) tokens shall be required for state-changing operations
SEC-023	API rate limiting shall be enforced (100 requests per minute per user)
SEC-024	File uploads shall be scanned for malware before storage
SEC-025	Content Security Policy headers shall be implemented
SEC-026	Security headers shall include X-Frame-Options, X-Content-Type-Options

11. System Integration

This section describes the external system integrations required for GRMTS operation. Note: Current implementation uses simulated integrations.

11.1 Integration Overview

External System	Purpose	Integration Type
NIA (Ghana Card)	Identity verification	REST API - real-time verification
GRA	TIN verification, tax reporting	REST API - real-time and batch
Lands Commission	Property ownership verification	REST API - real-time query
Ghana Post GPS	Digital address validation	REST API - real-time validation
MTN Mobile Money	Payment processing	REST API - real-time transactions
Vodafone Cash	Payment processing	REST API - real-time transactions
AirtelTigo Money	Payment processing	REST API - real-time transactions
SMS Gateway	Notifications, OTP delivery	REST API - real-time messaging
ECG/GRIDCo	Utility consumption data	Batch file transfer - weekly
Ghana Water	Utility consumption data	Batch file transfer - weekly

11.2 API Specifications

All external API integrations shall follow these standards:

- Protocol: HTTPS only (TLS 1.2+)
- Authentication: OAuth 2.0 or API Key in header
- Format: JSON for request and response payloads
- Timeouts: 30 seconds for real-time, 5 minutes for batch
- Retry: 3 attempts with exponential backoff for transient failures
- Logging: All API calls logged with correlation IDs
- Circuit Breaker: Implemented for high-volume integrations